UNIVERSITI TEKNOLOGI MARA

MALAY *PANTUN* THEME CLASSIFICATION USING SUPPORT VECTOR MACHINE

DANISH HAIKAL BIN SUHAIMI

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Malay *Pantun* Theme Classification Using Support Vector Machine

Danish Haikal Bin Suhaimi

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SUPERVISOR APPROVAL

MALAY *PANTUN* THEME CLASSIFICATION USING SUPPORT VECTOR MACHINE

By

DANISH HAIKAL BIN SUHAMI

2022646724

This thesis was prepared under the supervision of the project supervisor, Ts Dr. Hasiah Binti Mohamed @ Omar. It was submitted to the College of Computing, Informatics and Mathematics and was accepted in partial fulfilment of the requirements for the degree of Bachelor of Computer Science (Hons.).

Approved by

……

Ts. Dr. Hasiah Binti Mohamed @ Omar

Project Supervisor

JULY 13, 2025

STUDENT DECLARATION

I certify that this thesis and the project to which it refers is the product of my own work and that any idea or quotation from the work of other people, published or otherwise are fully acknowledged in accordance with the standard referring practices of the discipline.

……..

Danish Haikal Bin Suhaimi

2022646724

JULY 13, 2025

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ABSTRACT

This research focused on the study of Malay literary text, which is *Pantun* one of the Malay poetry form. The objective is to classify the Malay *Pantun* into its distinct types of themes as there is still lacks reliable prototype that classify the themes of Malay *Pantun* correctly with high performance. The Malay *Pantun* theme classifier implemented the Support Vector Machine algorithm into the model due to its robustness and effectiveness in text classification. The Malay *Pantun* dataset is manually collected from book, digitalized and processed to serve as the input of this research. The methodology phase of this research include preliminary analysis on the articles and journals related to implementation of Support Vector Machine in poetry text classification to gain a comprehensive understanding, processed data through preprocessing techniques such as text segmentation, case folding, tokenization, stopword removal, stemming and features extraction, designed phase by sketching the pseudocode, system architecture, flowchart and developed the Graphical User Interface (GUI), implementation of the Support Vector Machine algorithm into the developed prototype and training session and evaluation of the model performance using the testing data and result comparison with other parameters. The model is evaluated using the performance metrics such as accuracy, precision, recall and F1-score. The highest accuracy achieved by the model is 68.08%. Even though the result is lower than expected result, but it managed to classify the *Pantun* correctly into six types of distinguish themes. The model performance could be increased by integrating the model with the other algorithms or deep learning model.

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LIST OF ABBREVIATION

AI	Artificial Intelligence
AOA	Arithmetic Optimization Algorithm
BERT	Bidirectional Encoder Representations from Transformers
BOW	Bag of Word
DBP	Dewan Bahasa dan Pustaka
DT	Decision Tree
FN	False Negative
FP	False Positive
GA	Genetic Algorithm
GLCM	Gray Level Co-occurrence Matrix
GUI	Graphical User Interface
IEEE	Institute of Electrical and Electronics Engineers
K-NN	k-Nearest Neighbours
ML	Machine Learning
NB	Naive Bayes
NLP	Natural Language Processing
P	Input Variable
PSO	Particle Swarm Optimization
PSOGWO	Particle Swarm Optimization-Grey Wolf Optimizer
Q	Output Variable
RBF	Radial Basis Function
RF	Random Forest
RL	Reinforcement Learning
SL	Supervised Learning
SVM	Support Vector Machine
TF-IDF	Terms Frequency- Inverse Document Frequency
TN	True Negative
TP	True Positive
UNESCO	United Nations of Educational, Scientific and Cultural Organization

CHAPTER 1

INTRODUCTION

This chapter provides the background and rationale of the study. It give details of the significance of Malay *Pantun* to be preserved as the cultural heritage, the issues and problems that led to this research.

1.1 Background of Study

Machine learning has become an integral component of modern systems, enabling them to autonomously improve their performance through data-driven learning (Alpaydin, 2021). This advancement has revolutionized fields that rely on pattern recognition, including image processing, speech recognition, and natural language processing (NLP). NLP is one of the most domain core technology widely used to understand the natural language and the base of unique language particularly related on communication between human and the machine (Jin, 2022). Obviously, when the computer able to understand natural language from communication human and computer, it is possible for the computer to interpret the human language correctly. This includes the human cultural literary text such as poetry text classification, the growing area of study, which the classification by machine learning according to types of theme efficiently (Alawi et al., 2024). NLP plays a critical role in organizing and interpreting large volumes of textual data, offering insights that would be challenging to derive manually. The successful implementation of a poetry classification system not only gather a deeper insight in human literary for machine learning but also promote broader engagement with poetry and inspire future research in this area.

However, according to Pal & Patel (2020), most studies have primarily focused on major languages, such as English, which lead to a significant gap in researching the other country and races traditional literary text. Among these, Malay poetry also, have been affected by the gap leaving the study left a step behind. Malay poetry consists various of form such as *Syair*, *Gurindam*, *Sajak*, and *Pantun*. *Pantun*, a prominent form of Malay poetry, holds deep cultural and historical significance for the Malay community. Compared to other Malay poetic forms, *Pantun* is unique due to its concise structure, rhythmic patterns, and thematic variety, making it an essential expression of Malay identity and philosophy. Despite its importance in Malay cultural, efforts in performing the poetry classification are less compared to other language (Jamal et al., 2012). The available *Pantun* collections is in book published by Dewan Bahasa dan Pustaka (DBP) such as *Kumpulan Pantun Melayu* (Bakar, 2018). However, the dataset is only available in the physical books and not in the digital format. This has caused the dataset on the online source is limited. As a result, it becomes challenging for researchers, educators, and literary enthusiasts to identify the different categories of *Pantun* effectively.

This research aims to address these challenges by developing an automated classification system using Support Vector Machines (SVM). SVM is known for its robustness in handling high-dimensional spaces and its ability to classify complex data, making it a suitable match for tasks involving natural language processing, where word that have been vectorized and text pattern were analysed to classify the document accurately into its class. According to Irmanda & Astriratma (2020) results of the SVM algorithm as a classification method predicted *Pantun* with the highest results compared to other algorithms. By implementing the SVM in the classifier, *Pantun* can be classified into its distinct types enhanced the understanding and appreciation of this literary form and also help preserve it for future generations. Studies suggest that this poetic form continues to be relevant in shaping identity, despite concerns about its decline among younger generations due to the rise of modern influences and changes in communication practices.

1.2 Problem Statement

In this modern era, the classification of *Pantun*, a traditional form of Malay poetry, remains a challenge due to the lack of organised and accessible datasets. The data provided on the online source this day need to be labelled for efficient usage, otherwise the data without a proper labels is less utilized to it most and requires extra effort when needed (Pal & Patel, 2020). Based on the research conducted, it was found that the data is not systematically labelled according to its own types on online sources. This lead to misclassify of the *Pantun* into different category as the writer only labelled it according to their opinion. The standardised requirements to analyse and classify the *Pantun* into its respectively category is still ambiguous due to lack of regulation. Current existing collections of *Pantun* from various online sources do not classify poems into their specific types. These platforms tag content broadly as "*Pantun*" without distinguishing between the different *Pantun* categories. The classification of the document contribute vitally to information management and easily accessible which lead to a systematic labelling regulation to unlabelled documents (Pal & Patel, 2020).

Malay *Pantun* theme classification face a challenging task as there is lack available Malay *Pantun* dataset available on online source that have been well-categorised or tagged with its types of themes or classes. This study also face the limited well-organized dataset for this study area. Malay *Pantun* that published by the DBP only exists on physical books. To the best of my knowledge, lacks online dataset compiles these Malay *Pantun* data into a unified data collection except for the physical source such as books it is already classified into its types of themes. However, it still need to be extracted and scanned using the Optical Character Recognition (OCR) to be comprised into a single file. According to Irmanda & Astriratma (2020), the dataset used as input for Indonesia *Pantun* theme classification system is the data that have been properly labelled with their respective types of theme. As an effort, a collective dataset of Malay *Pantun* were collected from book of Malay *Pantun* would be

extracted and compiled into a single digital dataset for the study and contributed to others researchers that want to study on the Malay *Pantun* data structure.

Knowledge of the types of *Pantun* is still limited to experts in the field of literature, therefore a model is needed that can automatically classify a *Pantun* into which type (Irmanda & Astriratma, 2020). Since *Pantun* plays a significant role in Malay culture that covers a range of themes such as humour, love, advice, and tradition, it is essential to develop tools that make the classification process more efficient. Currently, researchers are forced to manually classify the poetry, which is time-consuming and limits further exploration. There are very limited research of the Malay literary text that available on the online database. It is figured out there is only an article that was studied on the Malay poetry classification system. However, the result obtained from the system by using SVM on average is 58.44%. The result of the accuracy was less satisfying for the poetry classification by theme but is well performed on predicting the text form is poetry documents or otherwise (Jamal et al., 2012). Therefore, this study aims to overcome these challenges and improve the current research by developing a classification system using SVM to automatically categorize Malay *Pantun* into its distinct types and achieve a better result than the previous articles.

1.3 Objectives

1. To identify the implementation of SVM algorithm in text and poetry classification and its features.
2. To develop a prototype of Malay *Pantun* theme classifier using SVM.
3. To evaluate the performance of the SVM algorithm for Malay *Pantun* theme classifier.

1.4 Scope

This study focuses exclusively on the classification of Malay *Pantun* using SVM. The aim is to develop a machine learning model that capable of automatically classifying *Pantun* into it distinguished themes.

1. **Target Users:** The Malay *Pantun* theme classification prototype is intended for students, enthusiast researchers, and cultural institutions with a focus on Malay literature. By automating *Pantun* theme classification, the system aims to assist these users in efficiently exploring and studying this literary form. Researchers can utilize the system to categorise *Pantun* into its themes, enhancing the research outcomes. Cultural institutions and educators may also get benefit by using the model as an educational tool to improve study experience and promote traditional Malay poetry among the youngsters.

2. **Dataset:** This project's dataset comprises *Pantun* is collected from the book published by DBP which is *Kumpulan Pantun Melayu* (Bakar, 2018). Each poem in the dataset is completely labelled to its respective type, spanning categories which are *Pantun Adat Resam* – Poems addressing customs and traditions, *Pantun Agama dan Kepercayaan* – religious and belief poems, *Pantun Budi* –gratitude, manners, and politeness poems, *Pantun Jenaka* – humorous and entertaining poems, *Pantun Teka-teki* – riddle-based poems, *Pantun Kepahlawanan* – heroism poems, *Pantun Nasihat dan Pendidikan* – advice and lessons poems, *Pantun Peribahasa dan Perbilangan* – idioms and numbers poems, *Pantun Kias dan Ibarat* – allegorical and parables poems, *Pantun Kembara dan Perantau* – travel and nomad poems and *Pantun Percintaan* – love poems. This dataset provides a structured foundation for the classification model and facilitated future research on Malay literature.

3. **Algorithm:** SVM is selected as the classification algorithm due to its accuracy and effectiveness in handling the text classification. Among others

algorithm, the SVM algorithm shown the highest average of accuracy when classify the poetry into its distinct category (Irmanda & Astriratma, 2020; Kaushal & Dutta, 2023). The SVM model were trained to categorize *Pantun* into eleven predefined types, such as *Pantun* Teka-teki (Riddle), *Pantun* Adat (Custom), and *Pantun* Agama (Religion) and others. Model performance is evaluated using metrics such as accuracy, precision, recall, and F1-score to ensure reliable classification (Alawi et al., 2024; Irmanda & Astriratma, 2020; Noh & Basri, 2021; Yang et al., 2024).

4. **Content:** The study is focused solely on the classification of Malay *Pantun* and does not cover other forms of Malay poetry, such as *Syair* or *Sajak*. The study also would be focused on the text analysis *Pantun* and would not include any other of input such as speech or poetry in form of image. The dataset is limited to publicly accessible sources, excluding dialect variations or modern adaptations of *Pantun* i.e. the mixed language of Malay and English *Pantun*. The system only able to receive a well-structured *Pantun* that follows the regulations form of *Pantun*. Additionally, due to time and resource constraints, the scope is limited to SVM without exploring more advanced algorithms. Despite these limitations, the project lays a foundational framework for future research in the automatic classification of Malay *Pantun.*

1.5 Significance of Study

This research contributed to the field of computational linguistic and cultural preservation by addressing the current challenges in organizing *Pantun*. The automatic *Pantun* theme classification play an important role in promoting the Malay culture heritage and preserved this identity to the younger generation. With the integration of the modern technology, a model that can classify the *Pantun* ensures the relevancy and originality of the *Pantun* remained accessible and guaranteed the preservation for future generations. One of the significance from the study is it overcome the research gap in AI implementation in Malay literary. As the study have been made, NLP has been implemented in a lot of

literature form for other language, but the focus on Malay poetry, which is the *Pantun* is very limited as there is only one paper that study this domain which is by Jamal et al., (2012). However, the result achieved is quite unsatisfying due to low result of performance achieved for the *Pantun* theme classification. Therefore, this study was aimed to overcome this challenge.

Next, the significance of this study is it provided a clear and well-organized structured Malay *Pantun* dataset for the future study. The prepared dataset available for all researcher and literature enthusiasts to be used for study. The dataset also organized and arranged according to its themes respectively. The collected dataset processed to ensure the efficiency of the Malay *Pantun* theme classification is able to deliver a satisfying result. Finally, the significance of this study is the automatic Malay *Pantun* theme classification model made working with *Pantun* in different context become easier. This model help to simplify the process of analysing and categorizing the *Pantun* whether for academic research, teaching or cultural activities. This effort ensure the heritage remained active as part of the cultural while preserving the Malay identity.

1.6 Research Outline

The framework outlined in Table 1.1 focuses on the development and evaluation of a Malay *Pantun* theme classification prototype using the Support Vector Machine (SVM) algorithm. It consists of five key phases: preliminary analysis, data preprocessing, design, implementation, and evaluations shown in Table 1.1.

Table 1.1 Research Outline

Objectives	Phase	Activity	Expected Deliverable
To identify the implementation of SVM algorithm in text and poetry classification and its features.	Preliminary Analysis	Preliminary Study	Comprehensive understanding of the SVM algorithm applied in text classification related to poetry and its features.
		Data collection	A consolidated Malay *Pantun* dataset.
To develop a prototype of Malay *Pantun* theme classifier using SVM algorithm.	Data Preprocessing	Process data	1. Well-organized dataset according to its types based on features and characteristics. 2. Processed Malay *Pantun* dataset.
		Data splitting	
	Design	Design flowchart, system architecture, pseudocode, GUI.	1. System flowchart. 2. System Architecture. 3. Malay *Pantun* theme classification pseudocode. 4. Template of Malay *Pantun* theme classification prototype GUI.
	Implementation	Prototype development and SVM implementation	A well-trained model that can classify the Malay *Pantun* into its distinct types using SVM classifier ready for user.
To evaluate the performance of the SVM algorithm for Malay *Pantun* theme classifier.	Evaluation	Asses the model performance.	Result of model performance based on: 1. Accuracy 2. Precision 3. Recall 4. F1-Score

1.7 Summary

This chapter underscore the role of machine learning in automating text classification, particularly in the scope of NLP, where it offers efficient classification of literary texts like *Pantun*. It identifies lacks study in research concerning literary text from other language, particularly Malay poetry, which includes various forms of poetry like *Pantun*. This chapter also highlights the challenges faced in the classification of Malay *Pantun*, primarily due to unorganized datasets and the manual effort of labelling the *Pantun* tends to mislabel, leading to inaccuracies in classification. To overcome the issues, the research proposes the implementation of SVM as the classifier aimed at accurately classifying Malay *Pantun* into predefined categories, thus enhancing both the understanding and appreciation of this cultural literary text.

CHAPTER 2

LITERATURE REVIEW

This chapter provided insight into the study and research that have been made on journal and articles to fulfil the first objective which to identify the implementation of SVM algorithm in text and poetry classification and its features. The literature review has been done using the search through online database such as IEEE, Scopus, MDPI and Google Scholar. Thus, a comprehensive understanding of SVM algorithms related to poetry classification were achieved.

2.1 Machine Learning

Humans have been innovating and using tools to improve and simplify their daily tasks since the beginning of the evolutionary era. Advanced machinery has been developed as a result of this inventiveness, especially in the area of artificial intelligence (AI), which meets a variety of human needs like computing, travel, and industrial automation (Mahesh that evaluate data and use the inputs to make well-informed predictions or decisions (Rincy and Gupta, 2020). Kang & Jameson (2018) explained that ML algorithms have been divided into four main categories which are supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning.

When labelled data is limited and unlabelled data is plentiful, semi-supervised learning integrate the aspects of supervised learning and unsupervised learning which is helpful (Thomas & Gupta, 2020; Mahesh, 2020). This method aims to reduce the effort required to manually label data while achieving effective data classification by using the small set of labelled data to classify or come out with patterns in the larger unlabelled dataset. On the other hand, labelled datasets are not necessary for unsupervised learning. Rather than

producing predetermined results, it finds hidden patterns and structures in the data (Mahesh, 2020; Gupta et al., 2022). This technique is frequently applied to reduce the dataset dimensionality or cluster data points into groups.

Finally, reinforcement learning uses a system based on rewards in which the algorithm gains decision-making skills through interaction with its surroundings. Rewards are given for successful outcomes, and penalties are given for unsuccessful ones. The machine can create strategies that optimise outcomes over time through this trial-and-error process (Gupta et al., 2022). Each of these three learning strategies has a distinct function in helping machines adjust, anticipate, and get better in a variety of situations. However Supervised learning is the most suitable ML category as the *Pantun* is included in the labelled dataset category and meet the characteristics of the supervised learning specification. Figure 2.1 shows the field of study in the ML method.

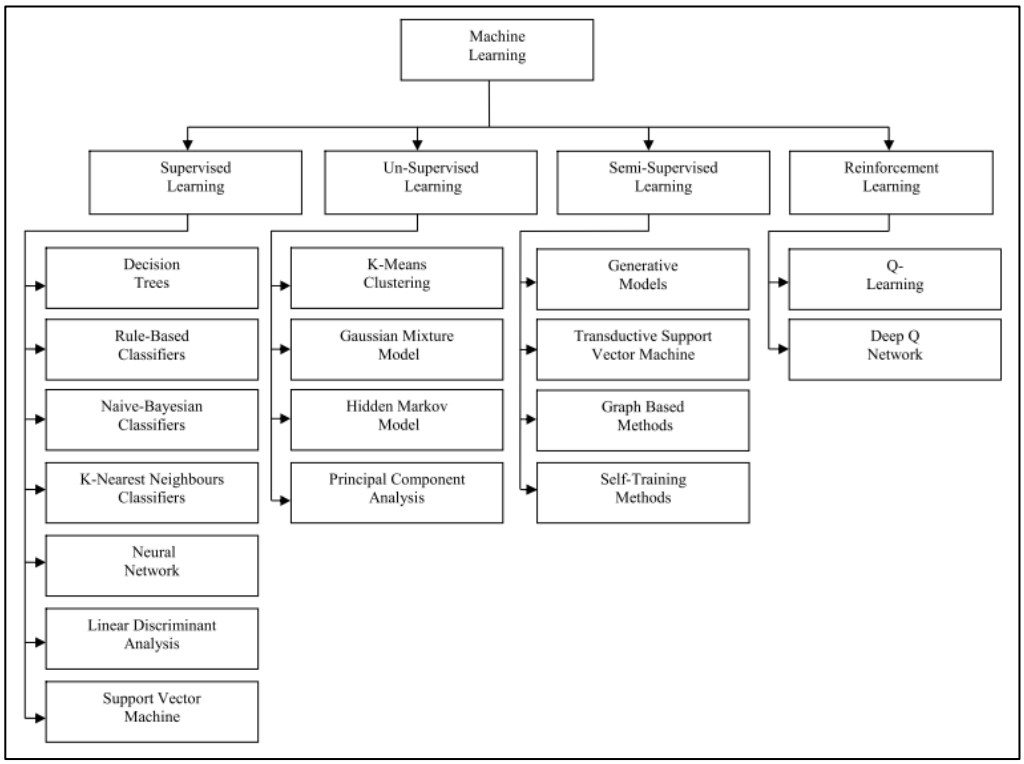


Figure 2.1 Field of Study in Machine Learning
(*Source*: Rincy & Gupta, 2020)

Table 2.1 shows the key points and sentences that have been simplified to give a clearer view on where the source of the citation is cited.

Table 2.1 Machine Learning Field of Study

References	Sentences
(Kang & Jameson, 2018)	ML algorithms is divided into four main categories which are supervised, unsupervised, semi-supervised, and reinforcement learning.
(Mahesh, 2019)	Implementation of machine in human life have made the task become easier in many aspects such as travelling, industries, and computing.
(Thomas & Gupta, 2020)	ML analyses the dataset, and the algorithms is implemented to make prediction using the provided input. It builds model that able to perform decision or prediction from the inputs.
(Thomas & Gupta, 2020; Mahesh, 2020)	Semi-Supervised learning learn from labelled data and to classify the unlabelled data. It combined both supervised learning and unsupervised learning. Supervised learning methods used properly labelled dataset that produced the predicted output.
(Mahesh, 2020; Gupta et al., 2022)	Unsupervised learning does not work under any supervision of teacher or return feedback. It have the input data but no output data. This helps the unsupervised learning to discover hidden patterns and insights in the provided data. It learn the patterns from the data and recognize the class of the data that holds same patterns which is widely used in feature reduction.
(Gupta et al., 2022)	ML learn from the data provided and the algorithm figured out what to do with the data. Once the algorithm have understood the dataset it worked automatically following the task assigned. Reinforcement Learning (RL). is based on return feedback from the prediction. If the algorithm give a good result it is rewarded with positive feedback otherwise, negative feedback.

2.2 Supervised Learning

A ML technique called supervised learning (SL) uses labelled dataset to train models for classification and predictive tasks. It uses labelled training data, where the input must be paired with an output in order to allow the techniques to learn mapping function between an input variable (P) and an output variable (Q) (Thomas & Gupta, 2020). By generalising patterns found in the training dataset, the goal is to develop a model that can reliably predict or classify the output for unseen inputs (Mahesh, 2020). The input dataset were split into two different subsets which are the training dataset that would be used during training session of the model and the other one is testing dataset which used to assess the performance during the evaluation phase (Mahesh, 2020). Labelled datasets, where the right answers are already known, serve as the main input and

guidance for supervised learning. When classifying fruit images, the model learns to associate input images that similar with the "apple" labelled image and if the input image is match it will produce the result as shown in Figure 2.2 (Gupta et al., 2022).

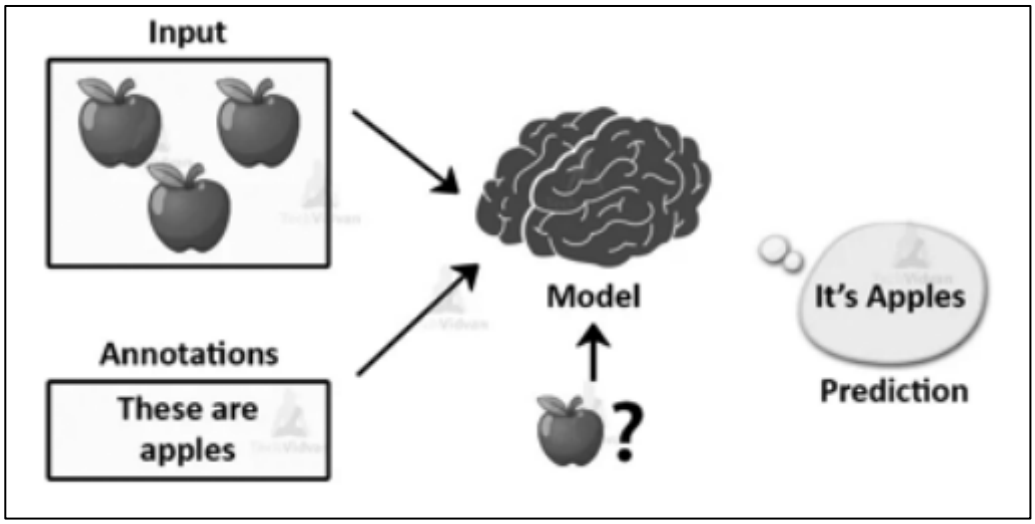


Figure 2.2 Supervised Learning Model
(Source: Gupta et al., 2022)

Regression and classification are the two main subcategories of supervised learning. Sorting data into predefined classes which may be binary or multi-class is the main goal of classification tasks. Usually, Multi-class classification tasks deals with more than two categories, like classifying objects as a car, dog, or tree, whereas binary classification involves only utmost two output, like deciding whether an email is spam or not (Gupta et al., 2022; Sharma et al., 2022). On the other hand, regression is concerned with forecasting continuous values and seeks to produce predictions that closely resemble the actual results. For example, it can forecast housing costs based on characteristics such as size and location, then the model accuracy is assessed by reducing the discrepancy between the expected and actual values (Sharma et al., 2022). Popular supervised learning algorithms that are made to handle various data types and problem scenarios include k-Nearest Neighbours (k-NN), Naive Bayes (NB), SVM, and Logistic Regression (Sharma et al., 2022). Numerous real-world applications, including fraud detection, medical diagnosis, and customer segmentation, are based on these algorithms. Table 2.2 shows the key

points and sentences that have been simplified to give a clearer view on where the source of the citation is cited.

Table 2.2 Supervised Learning Concept

References	Sentences
(Thomas & Gupta, 2020)	SL learn from the labelled training data. SL use an input variable (P) to produce an output variable (Q). SL studied the mapping function between Q and P which given the function is Q= f(P). The goal is studying the given training data which produced a complete function to the new instances .By using the function, the algorithm are able to correctly classify the instances from analysed and generalised labelled data
(Mahesh, 2020)	The function derived from the labelled dataset. The SL algorithms required external aids from this labelled dataset. The input dataset is divided into two different subset which are training dataset and testing dataset. The training dataset has Q which needs to be predicted or classified. Algorithm learned the pattern from this training dataset and then apply it on the testing dataset to evaluate the prediction or classification performance.
(Gupta et al., 2022)	Those data which already know the target answer is called label data sets. The collections of datasets of fruits are the input data. The name of input labelled data is apple. The model predict the input data is apple, so the output produced is correct. There are two different categories in SL which are regression and classification. The classification techniques consist of two types of data which are binary and multi. Binary class is used to divide the data in two outcomes or classes. The value is either 0 or 1. Example email received is Spam or not spam. Multi class is used to divide the data into 3 or more classes.
(Sharma et al., 2022)	Every output of SL has label or tags attached to it. The main aim is to find a specific class that is discretely associated in order to evaluate the correctness of the basis performed by the algorithm with given dataset
(Sharma et al., 2022)	SL anticipates a value that is closed to actual output from the given dataset and the evaluation is calculated based on the error value. The accuracy is considered high when the error value is small. Example of the SL algorithm are NB, KNN SVM and Logistic Regression.

2.3 Support Vector Machine (SVM)

SVM is a powerful SL algorithms commonly used in the regression and classification tasks. SVM was first created by Vapnik and his colleagues and is well known for its capacity to recognise a decision boundary known as the maximum margin hyperplane, which allows it to handle binary classification problems classes and also can be extend more than three classes (Patel et al., 2022; Chen et al., 2022). The best decision boundary is this hyperplane, which divides data points from various classes so that the margin or distance between

the closest points of opposing classes is as large as possible (Gupta et al., 2022). The model's accuracy and stability are guaranteed by the precise placement of these closest points, also known as support vectors. Maulidina et al. (2021) claim that SVM functions by projecting input of the data into a higher-dimensional space, where the use of a hyperplane facilitates class separation. The algorithm enhances generalisation and lowers classification errors by optimising the margin between each of the classes. Figure 2.3 shows the support vector machine apply hyperplane-margin concept.

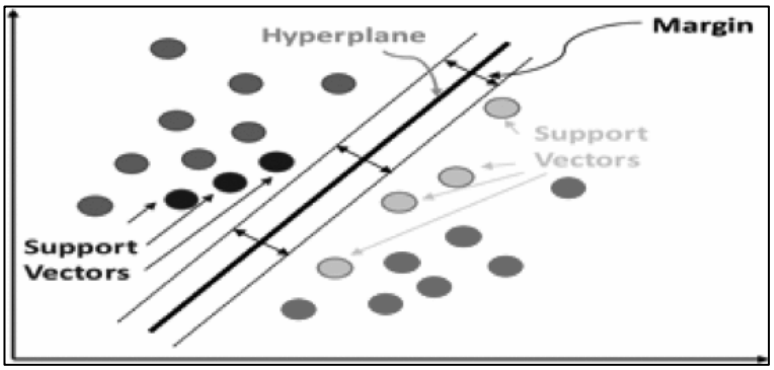


Figure 2.3 Support Vector Machine
(Source: Gupta et al., 2022)

The ability of SVM to solve and handle both linear and non-linear classification problems is one of its most noteworthy characteristics. SVM effectively converts data into higher-dimensional feature spaces by employing the "kernel trick," which makes it possible to classify intricate, non-linear datasets (Mahesh, 2020). Because of its resilience and versatility, the algorithm can be used in a variety of domain such as textual analysis, sentiment-based analysis, which categorises opinions as either positive or negative depending on evaluation characteristics (P.D & Haryoko, 2022). Furthermore, SVM's stability, superior generalisation skills, and versatility in processing data types make it a popular choice in a variety of study (Patel et al., 2022). SVM guarantees accurate and trustworthy predictions by reducing classification errors and optimising the margin between data classes. As demonstrated by its broad success across multiple domains, this makes it a perfect fit for tasks like text categorisation, image recognition, and even financial forecasting. Table 2.3 shows the key

points and sentences that have been simplified to give a clearer view on where the source of the citation is cited.

Table 2.3 SVM Concept

References	Sentences
(Mahesh, 2019)	SVM models is associated with learning algorithms that analyse data that have been used in classification and regression tasks. SVM performed the kernel trick on the non-linear classification to ensure efficient performance on linear classification. It draw margins between these classes. The closer the distance between margin and classes, the lower possibility of error in classification.
(Maulidina et al., 2021)	The input vector is mapped to the higher dimensional space where the distance between two parallel hyperplanes is separated by the maximum hyperplane separator.
(Chen et al., 2022)	SVM is a binary classifier for binary classification, and the maximum margin hyperplane are considered as the decision boundary.
(Patel et al., 2022)	SVM was developed by Vapnik and his colleagues and used for SL. It deals with problem of classification of two different classes and also can be extend more than three classes. SVM has been widely used in different area of study and shows the great performance.
(P.D & Haryoko, 2022)	SVM resilience and versatility allows it to be implemented in various problem solving such as superior generalization skills, stability, classification task, and processing linear and non-linear data proved SVM is a well-performed classification algorithm.
(Gupta et al., 2022)	The main purpose of the SVM is to allocate data in the best decision boundary & differentiates n space into classes. A new data point must be put into correct place. The best decision boundary is created from the hyperplane. The nearest positive points and the nearest negative points are called the support vector. The centre line is called the hyper plane and the distance between two parallel lines is called the margin.

2.4 Poetry

This section discussed on how poetry is defined based on the opinion from different authors in other cultures and languages. It also explained what is the types of poetry that exists in the Malay culture and society and the relevancy of the poetry to be research in the further study for the future generations and preserved as one of country identity, thus promote the *Pantun* among the youngsters. Lastly, cases study also were provided including the statistics to highlights the globally rate of poetry read by people in some society across the whole world from the author of National Endowment for the Arts website.

2.4.1 Authors Definition

Poetry is an art of cultural historical that have been inherited generation by generation among the citizens of the country around the world. Each country have its own definition and .aims to be delivered to the citizens. According to Jamal et al. (2012) poetry is a form of literary art in which widely used for its aesthetic and reminiscent high qualities in terms of the meaning. In other research from India, poetry is a piece of emotional-based literature text formed by inserting emotion and meaningful words through rhythm and thoughts (Kaushal and Dutta, 2024). Poetry is also a medium that use to express the feelings and emotions and convey concept through the rhythm and words (Sagar & Saini, 2024).

During ancient times, the poetry plays an important roles as an indirect advice from the senior generation, who have a lot of experiences and lesson learnt and deliver it to the younger generation. According to Ahmed et al. (2024), poetry plays an important role which it showed the culture of a society through their activities and beliefs which is form by their feelings and opinions. In Malaysia, specifically the Malay race, have their own traditional poetry which has various forms and types that differ one from another. It is widely use in the daily life as a forms of oral tradition such as in the opening ceremony by the emcee. Asmiaty (2024) highlights the word to be input need to be chosen carefully and wisely as it can carry a profound meaning.

Poetry is a traditional art of language in particular culture society which are used to communicate between each other but in a more pleasant and filtered way. It holds a cultural importance as it contains the advice to be delivered to the society as an indirect admonition including the high-status family. In traditional Malay society, they have a really high respect to their *Sultan*, so if anyone dispute the royal family, they would be punished. Nowadays, most poetry is delivered as an expression of confession, and some of them write the

poetry as their hobby. Table 2.4 show the gather information of poetry definition from different authors and the citation of the reference.

Table 2.4 Poetry Definition

References	Sentences
(Jamal et al.,2012)	Poetry is a form of literary art and used because of the unique structure and words contains aesthetics and vivid qualities in the meaning.
(Ahmed et al., 2023)	Poetry reflect cultural ideas and beliefs where People use poems as a way to express their feelings and self-opinions.
(Sagar & Saini, 2024)	Poetry is used as the medium to deliver feelings, and emotions.
(Kaushal & Dutta, 2024)	Poetry formed by adding emotional meaning to different words through the sentence component rhymes, rhythm or the meter.

2.4.2 Types of Malay Poetry

Malaysia is a country located in the southeast Asia, well-known for its multi-racial country that has been living in a peace for few centuries. Despite of this multiracial, Malay race is the origin people of this mother land. In conjunction, it holds a lot of culture symbolics and one of it is the traditional poetry. Malay traditional poetry can be break down into different types such as *Gurindam, Sajak, Syair and* also *Pantun. Pantun* is a form of Malay traditional poetry that built up from two, four, six, and up to fourteen lines (Amat Juhari, 2008). Furthermore,

Hassan (2020) underscores that a good poem that follows the rules and regulations of traditional Malay *Pantun* is mostly equal in words count and the rhythm ending with a/b/a/b in each verse. According to Ab Rahman et al. (2022), one of the *Pantun* advantage is, it shows the civilization and elegance in terms of common sense and the intellectuality of the traditional Malay community when they communicate with their surroundings. This can be seen when the Malay *Pantun* has a lot of nature elements inserted in the stanza. Table 2.5 shows the types of Malay poetry that still exists and inherit by the Malaysian society.

Table 2.5 Types of Existing Malay Poetry

References	Sentences
(Amat Juhari, 2008)	Malay poems exist in two, four, six, eight, ten, twelve lines, and fourteen lines
(Hassan, 2020)	A good poem must adhere to the syllables and rhymes a/b/a/b in the existing stanza.
(Ab Rahman et al., 2022)	one of the *Pantun* advantage is it reflects the common sense and the level of intellectual traditional Malay community utilizing the surrounding nature and implement it into their *Pantun* text.

2.4.3 Relevancy of The Poetry (*Pantun*)

Pantun, holds cultural importance as it has the esthetical values with the beauty of the language and the chosen words. This privilege of unique and creative ways of *Pantun* structure contributes on leveraging the Malaysia name and Malay culture into international level. For instance, The aesthetic of *Pantun* has been proven on December 17, 2020, where it was successfully announced as the winner of Intangible Cultural Heritage of Humanity category by the United Nations of Educational, Scientific and Cultural Organization (UNESCO) in France due to its aesthetics sentence structure (Ab Jabar et al.,2023). This achievement shows the *Pantun* has high values and being validated by the UNESCO that it is rich with beautiful of its language and its unique rhythm. So, as the *Pantun* is already approved in the eyes of the world, it must be preserved for the upcoming generation as one of the cultural that have been passed down by previous generations.

Pantun was introduced to the youngsters during their early age. Even so, students have difficulty in mastering the *Pantun* due to its unique structure and the words that been used in the stanza is too high (Mohamed et al., 2020). According to Ab Rahman et al. (2022), majority of the primary school students have less vocabulary knowledge, which giving them hard time to find the meaning and understand the *Pantun*. Furthermore, innovative and creativity must be a core in teachers as this is the fundamental to solve the problem appointed by student, because student acknowledge the teacher professionalism as someone who they can relied in academic (Zakaria et al.,

2021). However, Hamzah et al. (2019) clearly stated that teachers are preferred to follow their thinking cap, which some of them hesitate to teach the student about the *Pantun* or neglect the responsible regarding exposing the student to *Pantun*. Teachers must have undergone more appropriate teaching aids through courses on the Malay poetry especially the *Pantun*.

Nonetheless, the number of research that could contribute to *Pantun* teaching among the educator will remain limited (Ab Rahman et al., 2022). This lacks effort has cause bottleneck in explore the ML in the domain of Malay *Pantun.* Therefore, there is limited available automatic system regarding Malay *Pantun* that can assists in classifying the *Pantun* into it distinguish themes and category. There is limit in potential of research in *Pantun* domain among the educators and students as they manually classify the *Pantun* according to their unconfident instinct and different perspective despite a system that follows the specification and regulations of the classification which caused misclassification of the *Pantun*. Table 2.6 shows the relevancy in preserving the poetry as culture heritage based on the reference from different authors.

Table 2.6 Poetry Relevancy

References	Sentences
(Hamzah et al., 2019)	Teacher tends to neglect the importance of Malay *Pantun* and focus on other subtopics.
(Mohamed et al., 2020)	*Pantun* was introduced since primary school. However, students having hard time on mastering the *Pantun*.
(Zakaria et al., 2021)	Teacher must practice creativity and innovation in themselves to assists on guiding the students to create a high-quality *Pantun.*
(Ab Rahman et al., 2022)	Primary school students have limited exposure to *Pantun* and have limited knowledge of vocabulary.
	Primary school students have limited exposure to *Pantun* and have limited knowledge of vocabulary.
(Ab Jabar et al., 2023)	On December 17, 2020, *Pantun* was announced as the winner of Intangible Cultural Heritage of Humanity category by the United Nations of Educational, Scientific and Cultural Organization (UNESCO) in France.

2.4.4 Case Study of Poetry Classification

This study covered the classification of Malay poetry, which is specifically focus on the *Pantun*. Based on the statistics provided by Iyengar (2023), on 2022 data surveyed, 9.2% of adult access the poetry, which represents 22.4 million reader. Best to my knowledge, only a journal that have been made on the Malay poetry (Jamal et al., 2012). The study emphasize less efforts have been conducted on literary text such as poetry, specifically the *Pantun*. However, the results obtained from the research is below the expected results for the *Pantun* theme classification which is the highest accuracy obtained is 58.44%. Compared to the study that has been conducted by Irmanda & Astriratma, (2020), on classification of types of Indonesia *Pantun* using the SVM, the accuracy of the result obtained was above 81.71%. Table 2.7 shows a few case study that have been conducted related to poetry classification.

Table 2.7 Case Study of Poetry Classification

Author	Case Study/Statistics
(Jamal et al., 2012)	Limited research has been conducted on the literary text classification such as the poetry. The result of accuracy obtained by SVM classifier using Linear Kernel is 58.44%.
(Irmanda & Astriratma, 2020)	The results of accuracy by the SVM classifier to classify the *Pantun* into its themes is 81.91%.
(Iyengar, 2023)	2022 survey data shows 9.2% of adults, representing 22.4 million, read the poetry. It is lower than five years earlier in 2017.

2.5 Poetry Classification

This section discussed more details about the poetry classification which consists of the overview of poetry classification and classification system definition. It was highlighted that the study on poetry domain have been conducted from other languages such as Hindi and implementing the concept of machine learning and NLP together to classify poetry based on the features such as era, emotions and writing style. The classification of the textual analysis also rely on labelled data and feature extraction in order to deliver a reliable input.

2.5.1 Overview of Poetry Classification

In recent years, machine learning has improved on how poetry is classified into different genres. Pal & Patel (2020) stated on their research paper, an actual implemented experiment conducted to classify the Hindi poem into three distinguish class. A poem, one of the poetry type, was associated with a lot of information which it can be classified based on the writing schemes, the poets, compilation of emotions, and also the era of the poetry (Kaushal & Dulta, 2023). Monte-Serrate et al. (2021) emphasize the importance of the literary classification especially the poetry domain, can lead to a better search and enhance the accessibility of the digital data collections. In the other hand it is not only give a significant impact on information gathering, but demand of classifying different genres of the literary is high (Monte-Serrat et al., 2021). Table 2.8 shows poetry classification overview.

Table 2.8 Poetry Classification Overview

Author(s)	Sentence/Point
(Pal & Patel, 2020)	The classification of Hindi poem is classified into three classes which are Shringar, Karuna and Veera.
(Monte-Serrat et al., 2021)	A well-organized and classified literary according to their genres improved the web search experience and information gathering. Furthermore, distinguishing different literary genres is a high demand request from client.
(Kaushal & Dutta, 2023)	A poem is classified based on many basis such as writing schemes, basis of poets, emotions, era and etc.

2.5.2 Definition of Classification System

A text classification system is vital for categorizing poetry based on its structural elements. The classification system used supervised machine learning technique that required a set of data with labelled categories or classes to the given documents or stanza with the integration of NLP techniques (Kumar et al., 2022). Moreover, poetry classification is under the category of the text classification, and it is special to other regular prose text. According to Pal & Patel (2019), in the NLP, the two main components of the text classification are feature extraction and classification. Initially, the text must be transformed into a machine-readable numbers to allow the machine to interpret the text through feature extraction process which is submitted to the classification model (Pal & Patel, 2019). Table 2.9 shows the key points on definition of classification system from variety points of view of each author

Table 2.9 Definition of Classification System

Author(s)	Sentence/Point
(Pal and Patel, 2019)	Text classification have two main components which are feature extraction and classification. Feature extraction vectorizes the words into number form for the computer to read it and classify into predefine document categories.
(Praveen Kumar et al., 2022)	Text classification is a supervised machine learning technique that use labelled dataset that assigns a set of predefined categories or classes to the given text document with the combination of NLP techniques. Poetry is special text categories which is consider in literary text compared to other regular text.

2.6 Implementation of SVM Algorithm in Various Problem

SVM has been implemented in many domains in order to demonstrate the versatility and efficiency of the algorithm especially for classification tasks. For example, in the research that has been done by Chen et al. (2022), Arithmetic Optimization Algorithm (AOA) used SVM in identifying the chip that have the possibility of defected. The result of training accuracy is 94.23% while testing

accuracy is 84.88%.Moreover, Mohan & Jain (2020) covered for context of diabetes prediction, which evaluated the performance of SVM using various types of kernels. Their study highlighted that Radial Basis Function (RBF) kernel was performed better compared to other which hold the highest accuracy of 82% followed by Polynomial which is 80%, Linear, which is 77% and Sigmoid kernels which is 69%. The result obtained lead to new opportunities in the medical profession with SVM integration.

In other scope, Maulidina et al. (2021) also implemented the hybrid model of Particle Swarm Optimization (PSO) and Genetic Algorithm (GA) with SVM to classify the lung cancer into different types of stages. The result obtained from the hybrid model achieve accuracy of 97.69% compared to standard SVM which is 90.90% of accuracy. Sari et al. (2021) optimized the SVM and Gray Level Co-occurrence Matrix(GLCM) to classify the chili disease into their corresponding types. The research achieved an accuracy of 88%. Due to inconsistent light intensity, the result tends to misclassify the types of the disease. Though the research could be incorporate for additional features or hybrid techniques, the SVM method already successfully demonstrates the diagnostics potential in the agricultural.

In comparison, Jusman et al. (2021) also implementing a few ML techniques in the classification of Leukaemia cells using the Hu Moments features with SVM and K-NN. This study shows that SVM achieved a superior accuracy of 87.97% compared to K-NN which slightly left behind by the accuracy of 83.96%. Meanwhile, Hasanati et al. (2022) applied the SVM with a lexicon-based method for a better understanding of sentiment analysis on tweets regarding to COVID-19 vaccines. For the SVM parameter, RBF kernel was used to optimized the performance of the text classification. Thus, the result yields 85% of accuracy when applied the RBF kernel as the parameter. On the other hand, Noh & Basri (2021) compared the ML methods which are Random Forest (RF) and SVM in phishing websites detection. The result reveals that RF is more significant than SVM, with the accuracy of 99,98%, compared to SVM's

accuracy which is 84.73%. It was found out that for structured datasets such as website features, RF is the best method to handle it.

With the advancement of network technology, there also comes the intrusion in the network security. Chen et al. (2021) has developed a network intrusion detection system with optimized SVM parameters using Particle Swarm Optimization-Grey Wolf Optimizer-Support Vector Machine (PSOGWO-SVM). The accuracy this enhanced algorithm achieved is 94.4% followed by GWO-SVM, 90.4% and traditional SVM is 87.1%. SVM is also used across the sentiment analysis of online learning feedback, studied by Kurniawan & Wahyuni (2021), which classify the feedback of the students from the lectures class into positive and negative polarity. Their results showed overall accuracy is 84%. Lastly, the PSO parameter was applied in the SVM as an optimization techniques to classify the Go-Jek service review (P.D & Haryoko, 2022). They received the result from the model with the accuracy of 87% which was slightly better compared to their previous model that used GA as the parameter optimization for SVM, which is 86% of accuracy (P.D & Haryoko, 2022).

To conclude, the consistent of the SVM algorithm performance and its integrated model achieved the average accuracy above 80% through various domains highlight the robustness of SVM as the text classifier. The standard and standalone SVM is already obtained a good result but when it is hybridized with other techniques and methods it become more powerful and efficient. Thus, it is the best choice for this study to be conducted using SVM algorithms used to classify the Malay *Pantun* into different types. Table 2.10 shows the systematic review of the implementation of SVM in various tasks.

Table 2.10 Implementation of SVM Algorithm in Various Problem

No	Title	Year	Problems	Objectives	Algorithm	Result	References
1	Arithmetic optimization algorithm to optimize support vector machine for chip defect Identification	2022	Manual detection has fallen behind with the advancement of chip technology.	To identify and classify different types of chip surface defects using optimized Support Vector Machine.	AOA-SVM	The result obtained from the first set using AOA-SVM is 94.2308% on training set and 84.8837% on testing set while on the second set using SVM is 84.2308% on training set and 77.9070% on testing set.	(Chen et al., 2022)
2	Performance Analysis of Support Vector Machine in Diabetes Prediction	2020	Emergence of Diabetes among Indians is increasing due to lacks effective prediction system of the diabetes among the citizens.	To analyse the performance of SVM with different kernels. To determine the best kernel for diabetes prediction.	SVM	Four different Kernels of SVM is tested which are Linear, Polynomial, RBF and sigmoid. And RBF achieve the highest accuracy, which is 82%, while the rest Linear 77%, Polynomial 80% and Sigmoid 69%.	(Mohan & Jain, 2020)
3	Lung Cancer Classification using Support Vector Machine and Hybrid Particle Swarm Optimization-Genetic Algorithm	2021	Current method of cancer prevention inefficient due to error in observation process.	To Develop a model that can classify lung cancer precisely.	PSO-GA-SVM	The result obtained by PSO-GA-SVM is slightly higher compared to SVM without optimization which the accuracy of PSO-GA-SVM is 97.69% while SVM is 90.90% of accuracy	(Maulidina et al., 2021)
4	Classification of Chili Leaf Disease Using the Gray Level Co-occurrence Matrix (GLCM) and the Support Vector Machine (SVM) Methods	2021	Chili plants agricultural face losses due late detection of chili diseases such as Fusarium Wilt, Leaf Curl, and Yellow Virus.	To develop classification system that can diagnose and classify the types of chili disease.	-GLCM -SVM	The accuracy achieved in this research is 88% because some misclassification occurred due to intensity of light source.	(Sari et al., 2021)
5	Comparison between Support Vector Machine and K-Nearest Neighbor Algorithms for Leukaemia Images Classification using Shape Features	2021	Recognition of Leukaemia disease takes long time as it is difficult to differentiate between normal and Leukaemia cells.	To develop an image processing system that can classify blood cells using feature extraction.	-SVM -K-NN	The result of accuracy obtained by the SVM was 87.97% which is superior to K-NN 83.96%.	(Jusman et al., 2021)
6	Implementation of Support Vector Machine with Lexicon Based for Sentiment Analysis on Twitter	2022	Lacks sentiment classification that can classify the tweets regarding Covid-19 vaccines efficiently.	To classify sentiment from tweets using lexicon-based methods and SVM	-Lexicon-based -SVM	The result obtained by SVM with RBF Kernels is 85% of accuracy by tuning the parameter using grid search cross-validation.	(Hasanati et al., 2022)

7	Phishing Website Detection Using Random Forest and Support Vector Machine: A Comparison	2021	-Emergence of online phishing leading to scammed of user's personal information causing financial and data loss. -Variety structures of phishing cause difficulty to overcome control the issue.	To develop a system that can detect the phishing link in HTML code structure. To compare between RF and SVM.	-RF -SVM	The average accuracy result obtained by Random Forest is 99.88% while Support Vector Machine achieved 84.73%.	(Noh and Basri, 2021)
8	A Support Vector Machine with Particle Swarm Optimization Grey Wolf Optimizer for Network Intrusion Detection	2021	Current SVM method is inefficient to detect network intrusion due to randomly select the parameter and cause high rate of error.	To enhance the current network intrusion detection accuracy by optimizing SVM parameter using PSOGWO.	PSOGWO-SVM -GWO-SVM -SVM	The result of accuracy achieved by PSOGWO-SVM is 94.4% which is superior to GWO-SVM and SVM which is the accuracy is respectively 90.4% and 87.1%	(Chen et al., 2021)
9	Sentiment Analysis of Online Learning Students Feedback for Facing New Semester: A Support Vector Machine Approach	2021	Lacks automatic system to classify learning feedback based on the sentiment analysis, specifically into positive and negative responses.	To classify online class student feedback respectively based on sentiment analysis	SVM)	The result obtained was 84% of accuracy on sentiment analysis. SVM yields 77.61%, on the positive polarity, while the negative polarity value gets 94.26%.	(Kurniawan & Wahyuni, 2021)

2.7 Similar Works

The application of ML techniques is widely used in the literary classification which is in the scope of the poetry classification that consists of many algorithm and SVM is one of it. This show the ability of the SVM to adapt in diverse structures of literature language, which is quite unique and hard even for a human to interpret the real meaning of the poetry, but this algorithm are able to achieve a satisfactory result. In *Pantun* theme classification, Irmanda and Astriratma (2020) used SVM algorithms to classify the *Pantun* into three general categories and achieve an accuracy of 81.71%. This study used 470 data collected from book *Redaksi Balai Pustaka* which consists of 376 training data and 94 testing data.

Meanwhile, Bafna & Saini (2020) demonstrate that RF achieve a lower error rate compared to NB which is respectively 0.3 and 0.4 on the classification of Hindi poetry classification. The Scopus was retrieved from different websites and total data processed is 450 data. This result obtained shows that the ensemble methods like RF is more suitable to handle the complex datasets. In other study of Hindi poem's classification, conducted by Pal & Patel (2020), experimented multiple algorithms such as the SVM, Decision Tree (DT), RF, and NB. As the data is limited, the data was manually created which consists of more than 36000 data comprise Bengali poems. The result obtained from this study was superior to NB as it obtained the highest accuracy which is 52% compared to others which are SVM and K-NN are 44%, RF is 40% and DT is 32%.

In Punjabi poetry classification, Kaushal & Dutta (2023) emphasize SVM is superior other algorithm, when the algorithm achieved 82.17% of accuracy followed by RF, NB, and Logistic Regression. From various online resources, a total of 404 data were collected and divided into four different categories. Even though, the resources is limited, SVM model managed to outperform the effectiveness in text classification. For Bangla poem genre classification, Pasha et al. (2023) found that the NB achieved the highest accuracy with percentage of 86% among the ML model, outperforming Logistic

Regression and SVM respectively 84% and 83% of accuracy. Web scrapping techniques have been used to gather and collect 1131 of poems data from the "Banglarkobita" websites. This study underscores that even though SVM is performed wells, however selection dataset characteristics influence a significant impact to determine the final research final results.

Apart from that, Alawi et al., (2024) optimized Bidirectional Encoder Representations from Transformers (BERT) model for Arabic poetry classification and obtained result with accuracy of 86.82%. The data was fetched from Kaggle website and consists of 25000 records. Finally, Jamal et al. (2012) conducted study on the Malay poetry using SVM with Linear and RBF kernels. The dataset used in this is manually collected data from the book *Kumpulan Pantun Melayu* which consists of 1500 dataset. The result obtained on Linear Kernel is 58.44% and 100% for *Pantun* thematic classifying and recognized poetry form outperforming the RBF kernels. Although the result obtained for *Pantun* thematic classification was less satisfactory compared to other study, this study was marked as one of the earliest attempt to classify Malay Poetry computationally. However, there is very limited Malay *Pantun* dataset that available in the database especially the compiled versions.

In conclusion, SVM is reliable and effective because the consistency of the average performance through various study exceeding 80%. The consistency of SVM highlights the ability of adaptiveness is high especially when data set available is limited. This showed that it is really suitable for the usage for text classification which focused on the Malay *Pantun* theme classification as there is lacks research have been done on this domain and there were limited reliable resource of Malay *Pantun* dataset available on online resource. Table 2.11 shows the systematic review of the collection of similar work that related to the poetry classification using various ML algorithm and some of the paper used integration and optimization system with SVM to classify the text into its corresponding types and categories.

Table 2.11 Similar Work Related to Poetry Classification

No	Title	Year	Problem	Objectives	Algorithm	Dataset	Result	References
1	Klasifikasi Jenis *Pantun* dengan Metode Support Vector Machines	2020	Less study have been conducted on *Pantun* theme classification	To develop a model that can classify *Pantun* into different types.	SVM	Indo *Pantun* labelled dataset was retrieved from book *Redaksi Balai Pustaka* consists of 470 data which consists of 376 training data and 94 testing data.	The result obtained from this study SVM is able to classify types of *Pantun* with the accuracy 81.71%.	(Irmanda and Astriratma, 2020)
2	Poetry Classification Using Support Vector Machines	2012	Less effort have been done on Malay poetry classification because the Malay poetry holds broad and deep meaning yield in complex to interpret it.	To develop a model that can classify *Pantun* by theme.	SVM	The Malay *Pantun* dataset was collected from the book *Kumpulan Pantun Melayu* which consists total of 1500 dataset which divided into 10 different categories.	The result obtained using SVM using Linear Kernel with highest average percentage of 58.44% accuracy which is less satisfactory.	(Jamal et al., 2012)
3	Automatic Multiclass Document Classification of Hindi Poems using Machine Learning Techniques	2020	Underexplored domain as the Hindi Poetry structure is complex in multi sense nature of words, various sentence's structure and limited labelled dataset of Hindi poems.	To develop a system that can classify Hindi poems into three different Rassa.	RF, SVM, DT, K-NN, NB	As limited availability of the dataset, the dataset was manually created which consists of 36000+ verse which comprises Bengali poems.	Naïve Bayes achieved the highest accuracy with percentage of 86% while Logistic Regression and SVM respectively achieved 84% and 83% of accuracy.	(Pal & Patel, 2020)
4	Analysis of Performance Metrics for Classification of Punjabi Poetry using Machine Learning Techniques	2023	Limited resource and research focusing on Punjabi poetry genre classification	To evaluate the model performance using different algorithm for Punjabi poetry classification.	SVM, RF, NB, Logistic Regression	A total 404 of poems data collected from different online resources and divided into four different categories. http://www.punjabi-kavita.com http://punjabimaaboli.com	SVM is proven to be the best performing in terms of accuracy with the highest accuracy of 82.17% compared to another algorithm.	(Kaushal & Dutta, 2023)

5	Genre Classification of Bangla Poem Using Machine Learning and Deep Learning Techniques	2023	-Complex language and styles make the computational analysis becomes challenging. -Limited available dataset of Bangla poems.	To Enhance the poems genre classification accuracy using optimized ML and deep learning.	Logistic Regression , NB, SVM	1131 poems data collected from the "Banglarkobita" website using the web scrapping techniques. https://banglarkobita.com	For ML model, naïve bayes produced the highest accuracy, which is 86%, followed by Logistic Regression with 84% and SVM 83% of accuracy.	(Pasha et al., 2023)
6	BERT-AraPeotry: BERT-based Arabic Poems Classification Model	2024	Struggle to adapt in classifying the poem due to Arabic poetry classical language, complex structures and also usage of the metaphors.	To develop a model with sophisticated comprehension of the theme and poetry phrases in Arabic poems.	BERT	Dataset used in the research was 25000 record from the Kaggle websites "Arabic Poems Classification Challenge Dataset" https://kaggle.com/competitions/arabic-poem-classification	The model performance is effective as the Arabic poetry categorized with an accuracy of 86.82%.	(Alawi et al., 2024)
7	Hindi Poetry Classification using Eager Supervised Machine Learning Algorithms	2020	Less research on Hindi text compared to English as the Hindi text is complicated and lacks resources availability of Hindi text.	To develop a model that can classify the Hindi poetry text into its corresponding category.	NB, RF	The corpus dataset was retrieved from different websites and total data processed is 450 data. https://aajtak.intoday.in/sahitya-kavita.html	Random forest techniques achieved less error which is 0.3 depicts good classification compared to Naïve Bayes which is 0.4.	(Bafna & Saini, 2020)

2.8 Implication

After the comprehensive review of the study, ML algorithm such as NB, RF, GA, and K-NN is widely used in a lot of research paper and SVM is also most frequently used algorithm. This highlights that the algorithm of ML is able to adapt and deal with many problem in various area of domain including text classification. The effectiveness and versality of SVM is the most performed algorithm to be used in the text classification tasks. SVM has consistent performance, which is the result yielding average above 80%, simultaneously emphasize its reliability and adaptability to complex datasets especially in the literary domain. When the SVM is hybrid with other techniques such as PSO, GA, AOA, SVM's performed much effectively as demonstrated by Chen et al. (2022), Maulidina et al. (2021) and P.D & Haryoko (2022). Maximum accuracies were achieved and increased computational efficiency, making the SVM the best choice for complex classification tasks.

SVM has been proven effective as a reliable algorithm for poetry classification through various studies. Much research have been made on different languages of poetry such as Hindi, Punjabi, Bangla, Arabic and also a few for Malay poetry represent adaptability of the SVM in handling the linguistic rich and structurally diverse datasets. Algorithm such as NB and RF have been outperformed by SVM algorithm for Punjabi poetry classification by achieving 82.17% accuracy (Kaushal & Dutta, 2023). SVM constantly tried to deliver competitive result in order to proof it reliability even though in some case NB have slightly higher accuracy than SVM such as Bangla poetry classification (Pasha et al., 2023). To be considered, Jamal et al. (2012), conducted an early experiment and study on Malay poetry classification using SVM with Linear and RBF kernels. However, the result of accuracy obtained was quite lower compared to modern study, but it can be the benchmark paper for Malay *Pantun* computational analysis.

Based on the studies and the performance of the SVM, SVM is the best algorithm for Malay *Pantun* theme classification. It is a perfect choice for

classifications tasks especially text classification because it consistently delivers reliable results, mostly above 80% accuracy. It is a good decision as SVM has been test across various sorts of data including poetry classification in different languages. SVM also have been used to handle language with unique structures like Hindi and Punjabi. For some case, even SVM is not the top performer, it still gave a competitive results with the other algorithm. As SVM is reliable and flexible, it worked well to handle mild complex linguistic features of Malay *Pantun* which is related to the project. It is also have been used in previous research, but the result is not satisfactory, so an enhancement can be made on the existing system and a better result were achieved. Due to lacks availability of dataset Malay *Pantun* and limited research resources, this study provided a great opportunity to contribute significantly to linguistic and cultural preservation. Moreover, the success of the Malay *Pantun* theme classification prototype contributed to preserving and understanding Malay Poetry deeper.

2.9 Literature Review Summary

Malay *Pantun* theme classification is one of the domain of study that applied the AI in the classification task. The AI field used in the *Pantun* theme classification is from the combination of ML algorithm and also NLP process. There are various algorithm that are able to handle the classification task such as SVM, NB, K-NN, and Logistic Regression. Further study also conducted on Malay poetry to identify types of Malay poetry form and the structure of the *Pantun*. Through this, the best algorithm that were used in this research is SVM due to its robustness, effectiveness and the ability to adapt with the literary text classification that contains unique structure such as Malay *Pantun*. Figure 2.4 shows the spider map of the literature review.

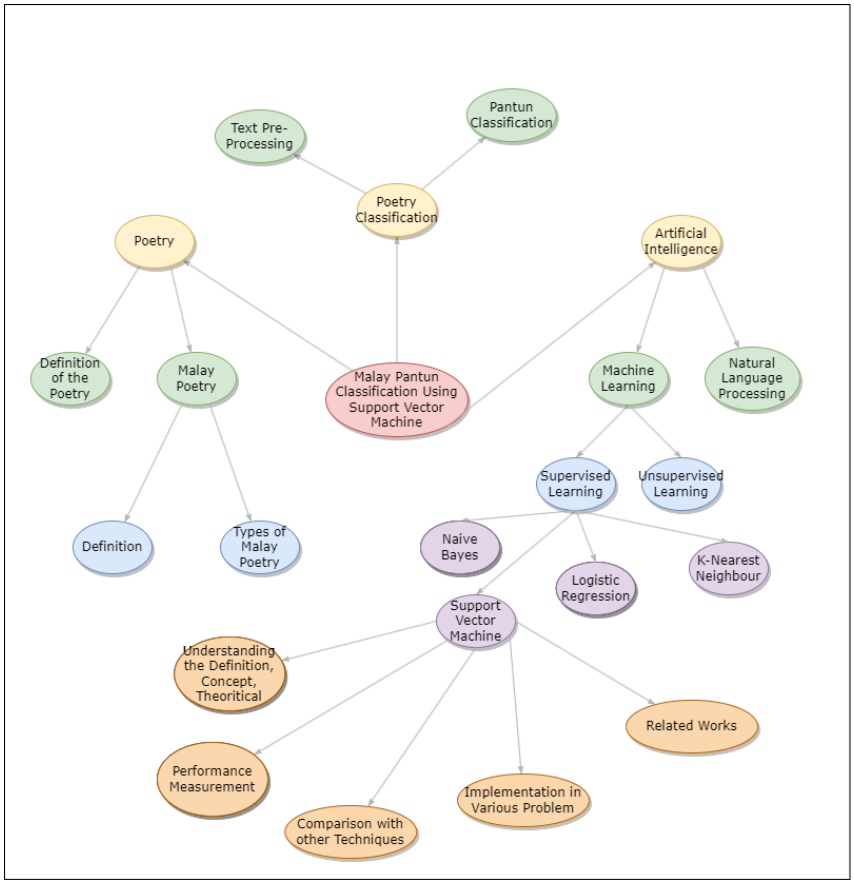


Figure 2.3 Literature Review Spider Map

2.10 Conclusion

To conclude, this chapter discussed about the literature review to achieved the first objective of the project which is to identify the use of SVM algorithm in poetry classification. The aim of this objective is to gain a deeper and comprehensive understanding and knowledge on the three major scope, which are field of study related to poetry, ML methods , and the implementation of SVM algorithm in the various problem and similar work related to poetry classification. Through this review, it was found that SVM is the most reliable and perform well for classification task, especially on text classification as it achieved accuracy that outperforms most of the other algorithm and the ability to handle various problem-solving tasks.

CHAPTER 3

METHODOLOGY

This chapter discussed about the details for each methodology phases which include preliminary study, design, implementation and evaluation. Several activities were done to achieve the objectives. The details of the activities is explained in the task specification. Some tasks require tools and method during the research period to gain information and ensure the expected deliverables are achievable

3.1 Overview of Research Methodology Framework

This section discussed about the details of the methodology phases that applied from Bafna & Saini (2020) and Kaushal & Dutta (2023) that consists of preliminary analysis, data preprocessing, design, implementation and evaluation. Each phase have its own objectives or shared a single objectives with consecutive phases. Activities conducted precisely with detail tasks to achieve the objective. Each phase have expected deliverables that gain at the end of the phase. Preliminary analysis covered the research finding, information gathering and data retrieval. Data preprocessing discussed on how the raw data is processed to be a well-organized structured data. Design phase discussed on how the GUI design and system design using the Python framework. Implementation phase focused more details on prototype development, training the model and implemented the model. Final phase, which is evaluation phase assessed the performance of the model based on the metrics. Table 3.1 shows the research methodology framework.

Table 3.1 Research Methodology

Objective	Phase	Activity	Method / Tools	Expected Deliverable
To identify the implementation of SVM algorithm in text and poetry classification and its features.	Preliminary Analysis	**Activity :** Preliminary Study **Task :** • Review literature on SVM algorithms, text classification and evaluation metrics. • Search the articles and journal through online database	Online databases such as IEEE, MDPI, Scopus and Google Scholar.	A detailed explanation on the implementation of the SVM algorithm used in text classification related to poetry and its features.
		Activity : Data collection **Task :** • Collecting the *Pantun* data from *Pantun* book.	Source : DBP book called *Kumpulan Pantun Melayu*	A consolidated Malay *Pantun* dataset.
To develop a prototype of Malay *Pantun* theme classifier using SVM algorithm.	Data Preprocessing	**Activity :** Process data **Task:** • Identify and amend the erroneous records within the dataset. • Performed data cleaning and preprocessing techniques	Tool: Python, Microsoft Excel Method: Text segmentation, case folding, tokenization , stopword removal, stemming, feature extraction.	1. Well-organized dataset according to its types based on features and characteristics. 2. Processed Malay *Pantun* dataset.
		Activity : Data splitting **Task :** Separate data into training and testing data consist of three ratio. • 90%-10% • 80%-20% • 70%-30%	Tool: Python Method: stratified sampling.	
	Design	**Activity :** Design **Task :** • Design the flowchart. • Design the system architecture. • Design the pseudocode. • Design the GUI.	Tool: Draw.io, Canva.	1. System flowchart. 2. System Architecture. 3. Malay *Pantun* theme classification pseudocode. 4. Template of Malay *Pantun* theme classification prototype GUI.
	Implementation	**Activity :** Implementation **Task :** • Create a prototype using the designed prototype • Implement the SVM algorithm using Python. • Train the model using training dataset.	Tool : Python.	A well-trained model that can classify the Malay *Pantun* into its distinct types using SVM classifier ready for user.
To evaluate the performance of the SVM algorithm for Malay *Pantun* theme classifier.	Evaluation	**Activity :** Evaluate prototype **Task :** • Perform test to assess the model performance. • Compare the result of performance metrics using different parameters.	Tool: Python Method: • Performance metrics: such as accuracy, precision, recall, and F1-score • Cross-validation	Result of model performance based on: 1. Accuracy 2. Precision 3. Recall 4. F1-Score

3.1.1 Details Description of The Research Methodology

Initially, the research methodology begun with the preliminary study. The objective of the preliminary analysis is to identify the implementation of SVM algorithm in text and poetry classification and its features. A literature review conducted found out SVM is one of the Machine Learning (ML) algorithms widely used in text classification such as the poetry classification (Irmanda & Astriratma, 2020; Jamal et. al., 2012), sentiment analysis(Kurniawan & Wahyuni, 2021) and classification of the Go-Jek service review (P.D & Haryoko, 2022). A reliable dataset were also collected from published *Pantun* book. The data were collected through DBP book called *Kumpulan Pantun Melayu* (Bakar, 2018). The dataset of Malay *Pantun* were identified as the expected deliverables.

In the data preprocessing phase, the data collected were processed. The data preprocessing is a significant process because the *Pantun* contains rhymes and linguistics features that need to be extract which is forwarded to the model for the classification (Alawi et al., 2024). Malay *Pantun* consist of lot of linguistics style in the writing, and this affect the model as it unable to interpret the real meaning of the human language because it only understand the computer language which in numerical format. Moreover, the *Pantun* text must be processed to extract the real meaning, which determine the themes of the *Pantun*, normally reside in the second part of the verse, usually in third and fourth line for four verse *Pantun* structure.

According to Irmanda & Astriratma (2020) text preprocessing include text segmentation, case folding, tokenization, stopword removal and stemming the word that convert word to the root word. A processed and well-organized Malay *Pantun* dataset is collected into single file. Next, the data split into training and testing data consist of 3 distinct set which are 70% - 30% referred to research of Jamal et al. (2012), 80% - 20% by the study of Irmanda & Astriratma (2020), and 90% - 10%. The model was trained using these three

distinct set of training Malay *Pantun* dataset to identify the ratio that contribute the best performance result for the SVM Malay *Pantun* theme classifier model.

After that, the design phase carry on with design of the user interface for the prototype. The prototype interface developed using the Python framework in order to give the user a platform to interact with the model and test the model later. Potentially, an interface of the Malay *Pantun* theme classifier prototype developed by the end of the phase. The system architecture and flowchart and the pseudocode of the model also were designed as a guidance during the development phase. The implementation phase begun with developing the classifier model. The classifier model is implemented with SVM algorithm. A model that able to handle the *Pantun* theme classification was successfully produced. The model was trained using the training dataset. The model able to classify the Malay *Pantun* precisely into its corresponding types of themes. These consecutive phases are aimed for the second objective which is to develop a prototype of the Malay *Pantun* theme classifier using SVM algorithm.

Finally, the evaluation phase focused on the assessment of the SVM model performance. The model with the best performance tested with testing dataset. The metrics used are accuracy, precision, recall and F1-score (Alawi et al., 2024; Irmanda & Astriratma, 2020; Kaushal & Dutta, 2023; Pasha et al., 2023). Mazlin (2022) stated that the accuracy metrics were used for classification assessment because it is the simplest method, which the score given by number of correctly classified sample. Precision is where the number of correctness the documents predicted while recall are measured by prediction of document assigned correctly to it documents dataset, and F1-score is used to measure overall model's performance by calculating the average of precision and recall. By running the system on the user interface, the result of the metrics involved were achieved with the best performance.

3.2 Preliminary Analysis

Table 3.2 shows the preliminary study phase that consists of literature review and data collection.

Table 3.2 Preliminary Study Phase

Objectives	Phase	Activity	Method/ Tools	Expected Deliverables
To identify the implementation of SVM algorithm in text and poetry classification and its features.	Preliminary Analysis	**Activity :** Preliminary Study **Task :** • Review literature on SVM algorithms, and text classification. • Search the articles and journal through online database	Online databases such as IEEE, MDPI, Scopus and Google Scholar.	A detailed explanation on the implementation of the SVM algorithm applied in text classification related to poetry and its features.
		Activity : Data collection **Task :** Collecting the *Pantun* data from *Pantun* book.	Source : DBP book called *Kumpulan Pantun Melayu*	A consolidated Malay *Pantun* dataset.

3.2.1 Preliminary Study

The research finding has been conducted for three major domain which are the ML, SVM, and the implementation of SVM in the poetry classification system. The aim of the study was to achieve a comprehensive understanding of the basic foundation of the SVM algorithm. Throughout the research, SVM is one of the most popular algorithms used for text classification including the poetry text classification. ML is used widely in the modern era to fulfil the human technology requirements such in the industrial automation, computing and also the travel (Mahesh, 2019). ML enable the machine to learn from the historical data and automatically enhance the performance without a requirement of definite programming (Gupta et al., 2022). According to Kang & Jameson (2018), ML is divided into four main categories and one of it is the Supervised Learning(SL).

SL uses labelled data as input variable in order to train models for text classification and prediction tasks yielding the output variable. The

corresponding output generated the mapping function (Rincy & Gupta, 2020). The mapping function pattern was generalized to allow the model becomes reliable for predicting a novel output using unseen inputs (Mahesh, 2019). Furthermore, the input used is training dataset for model development, and test dataset for the evaluation phase. Regression and classification are two main subcategories of SL. Regression is forecasting and seeking prediction product closed to actual results while classification is sorting the data into predefined class (Gupta et al., 2022; Sharma et al., 2022). SL has many popular algorithms and one of the most reliable algorithms for classification tasks is SVM.

SVM is powerful algorithms used for classification tasks. It was introduced and designed by Vapnik and his colleagues and well-known for the ability to recognise boundaries known as the margin hyperplane, allows the binary classification problem to be handled (Patel et al., 2022; Chen et al., 2022). SVM divided various classes, so that distance between closest points in class is maximized (Gupta et al., 2022). Precision placement of these closest points, also known as support vectors, it enhanced the model's accuracy and stability. SVM converts data into higher-dimensional space by using "kernel trick," which allows complex classification of non-linear datasets (Mahesh, 2019). Thus, its versatility on solving various problems across multiple domains fit perfectly for text classification tasks.

3.2.2 Data Collection

Data is one of the vital core on making the algorithms of ML works efficiently especially for the Supervised Learning such as Support Vector Machine (SVM) algorithms. However, it was found out that there is lack of digitalized Malay *Pantun* dataset that is available on the online platform. The closest one to the Malay *Pantun* that is available is only Indonesia *Pantun* which available on Kaggle websites. Therefore, the data is collected from the books published by DBP which is *Kumpulan Pantun Melayu* (Bakar, 2018).There 2052 dataset

have been collected. The data collected only for the Malay *Pantun* itself with predefined category.

Malay *Pantun* collected consists of eleven distinct types of themes and each of the themes have different number of *Pantun*. In some themes, there are stanza that build from two, four, six and eight lines. Each *Pantun*, have two which the first part is the indicator, and the second part is the real meaning. The indicator and the real meaning is differentiated with the delimiter. The *Pantun* used for the project is the stanza that build up from the four lines. The Malay *Pantun* were selected from six different themes. Each of these poems consist of 120 Malay *Pantun* which have the sum total of 720 *Pantun*. The 120 *Pantun* data were chosen because it represents the highest count of *Pantun* available across all six themes. Figure 3.1 shows the sample of the dataset Malay *Pantun* that have been collected.

Lima jari batang berduri, Masak gula buat manisan; Saya datang kerana budi, Bertambah pula tuan berpesan.	PANTUN BUDI
Saya tidak menanam nanas, Tanam kepaya di dalam padi; Saya tidak memandang emas, Budi bahasa yang saya cari.	PANTUN BUDI
Sebab padi sebab selasih, Itulah sebab saya lurutkan; Sebab budi sebab kasih, Itulah sebab saya turutkan.	PANTUN BUDI
Semenjak kentang yang jadi gulai, Ubi tidak bersama lagi; Semenjak wang jadi pemakai, Budi tidak berguna lagi.	PANTUN BUDI
Pergi ke pulau menggali ubi, Singgah ke beting dilayar juga; Sepuluh kali kita berbudi, Kalau miskin terbuang juga.	PANTUN BUDI
Lazat pula cita rasanya, Jeruju dengan durinya; Serta dengan budi bahasanya, Setuju pula dengan isterinya.	PANTUN BUDI
Pada berlari baik berlompat, Barang dicari segera dapat; Sudah berbudi jangan diumpat, Balasan baik syurga dapat.	PANTUN BUDI
Buah lada lebat di pangkal, Gugur daun dengan buahnya; Sungguh ada ringgit berkapal, Kalau tak berbudi apa gunanya.	PANTUN BUDI
Anak sulung dalam telaga, Mari kukait mata rahat; Takdak untung badan celaka, Buat baik jadi jahat.	PANTUN BUDI
Buat ketupat bertali-tali, Mana tergantung sudah berisi; Tak tengok tempat mahu berbudi, Sangkakan untung berjumpa rugi.	PANTUN BUDI
Bintang timur tinggi, Jatuh julai nipah; Abang duduk mengaji, Adik minta sepah.	PANTUN JENAKA DAN PERMAINAN
Pisang kelat Jambi, Tanam balik rumah; Abang lambat mari, Sirih lipat habis punah.	PANTUN JENAKA DAN PERMAINAN
Beli rojak pergi ke pekan, Ke pekan Labis beli selipar, Ada budak merajuk makan, Nasi habis perutnya lapar.	PANTUN JENAKA DAN PERMAINAN
Laksamana pergi mendekut, Dapat seekor punai gading; Adakah singa memakan rumput? Adakah lembu memakan daging?	PANTUN JENAKA DAN PERMAINAN
Cekur udang kamat, Gali cekur tanam kunyit; Adik di rumah gamat, Nasi di dapur habis hangit.	PANTUN JENAKA DAN PERMAINAN
Ndoi-ndoi cak, Cak makan padi; Adik duduk teriak, Emak ketam padi.	PANTUN JENAKA DAN PERMAINAN
Hujan renyai-renyai, Pacat makan kaki; Adik hendak berinai, Abang nak berlaki.	PANTUN JENAKA DAN PERMAINAN

Figure 3.1 Sample of Malay Pantun Dataset

3.3 Data Preprocessing

Table 3.3 shows the design phase which includes the data processing and data splitting.

Table 3.3 Data Preprocessing Phase

Objectives	Phase	Activity	Method/Tools	Expected Deliverables
To develop a prototype of Malay *Pantun* theme classifier using SVM algorithm.	Data Preprocessing	**Activity :** Process data **Task:** • Identify and amend the erroneous records within the dataset. • Performed data cleaning and preprocessing techniques.	Tool: Python, Microsoft Excel Method: Feature extraction, text segmentation, case folding, tokenization , stopword removal, stemming, feature extraction.	1. Well-organized dataset according to its types based on features and characteristics. 2. Processed Malay *Pantun* dataset.
		Activity : Data splitting **Task :** Separate data into training and testing data consist of three ratio. • 90%-10% • 80%-20% • 70%-30%	Tool: Python Method: stratified sampling	

3.3.1 Preprocessing Stages

Processed data is significant as the data were used as the input to train the model. It is important to ensure the Malay *Pantun* data is cleaned and reliable, so the output produced by the model is correct. Preprocessing data steps involved text segmentation, case folding, tokenization, stopword removal stemming and features extraction. Figure 3.2 shows the flow of data preprocessing for *Pantun* text classification.

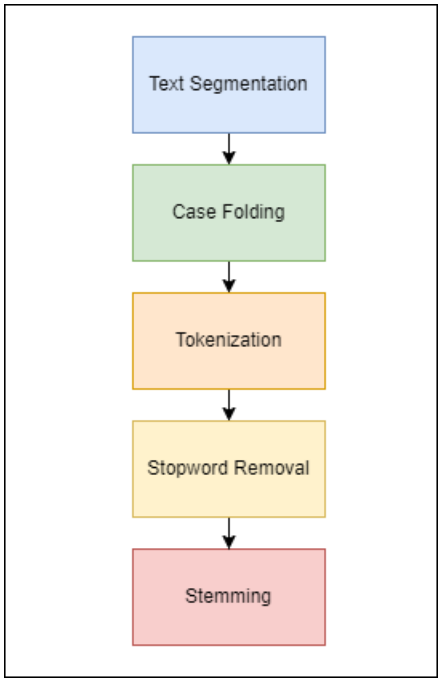


Figure 3.2 Data Preprocessing Flow

a) Text Segmentation

Malay *Pantun* consists of two major part which is the first half is the indicator of what is coming, and another half is the message or real meaning of the *Pantun* (Ahmad, 2002). Therefore, text Segmentation is needed for break the whole stanza of *Pantun*. The significant on the *Pantun* theme classification is reside on the meaning part, so it is required only the second half. Table 3.4 shows the text segmentation stage.

Table 3.4 Text Segmentation

Before Text Segmentation	After Text Segmentation
Pisang emas dibawa belayar,	
Masak sebiji di atas peti;	
Hutang emas boleh dibayar,	Hutang emas boleh dibayar,
Hutang budi di bawa mati	Hutang budi di bawa mati

b) Case Folding

After the stanza is break into the meaning, then the case folding step is applied where each word is converted to lowercase. All word is

standardized to ensure it treated evenly. In this step, special characters such as punctuation mark are being removed. Table 3.5 shows the case folding stage.

Table 3.5 Case Folding

Before Case Folding	After Case Folding
Hutang emas boleh dibayar,	hutang emas boleh dibayar,
Hutang budi di bawa mati	hutang budi di bawa mati

c) Tokenization

Tokenization applied to each lines where sentence were broken into a smaller part of subunits words whenever a white space is met (Grefenstette, 1999). The data is processed into three operations which document conversion to word counts which is bag of word (BOW), removing empty white space, and segmented input into a list of features which are also called tokens. Table 3.6 shows the tokenization stage.

Table 3.6 Tokenization

Before Tokenization	After Tokenization
hutang emas boleh dibayar, hutang budi di bawa mati	hutang
	emas
	boleh
	dibayar,
	hutang
	budi
	di
	bawa
	mati

d) Stopword Removal

The stop word removed from the lines. Stopword is a list of word that is commonly repeated in the text document such as the conjunction or pronouns. It does not significantly influence the *Pantun* actual meaning. Other than that, a word that frequently appears in the text is considered to

be removed. This step is ensuring efficiency of NLP task such as text classification. Table 3.7 shows the stopword removal stage.

Table 3.7 Stopword Visualization

Before Stopword Removal	After Stopword Removal
hutang	hutang
emas	emas
boleh	boleh
dibayar,	dibayar
hutang	hutang
budi	budi
di	
bawa	bawa
mati	mati

e) Stemming

Stemming involved convert word back to its root word form. Some words has affix, either prefix, infix or postfix. The process is used to reduce the features space number and enhance the performance of the classifier when the different form of features are stemmed into single features (Grefenstette, 1999). Therefore, it is only required the root word in order to train the model on interpret the meaning of the *Pantun*. Table 3.8 shows the stemming stage.

Table 3.8 Stemming Visualization

Before Stemming	After Stemming
hutang	hutang
emas	emas
boleh	boleh
dibayar,	bayar
hutang	hutang
budi	budi
bawa	bawa
mati	mati

3.3.2 Features Extraction

After the processing step is completed, the data features and characteristics were extracted. In this stage, the concept of TF-IDF is implemented in order to calculate the frequency of each stanza document and the whole document. As the machine are not able to understand and interpret human language, so TF-IDF one of the most widely known and weighting method is used to give vectorization to the root words which enable machine to understand the word in numbering format (Ahmed et al., 2022). A document collection , D , a word, w, and stanza document, d. The weight is calculated using (1) and (2) (Ramya & Pinakas, 2014).

$$TF = TF \times IDF \tag{1}$$

$$Wd = fwd \; x \; log \; (D/fwd) \tag{2}$$

Term Frequency (TF) is number of times *Pantun* word, w, which appears in a stanza document, d, while the Inverse Document Frequency (IDF) is number of times *Pantun* word, w, which appears in the whole document, D. The result of the TF-IDF is vector with various term along their term weight.

3.3.3 Data Splitting

After the data is processed, the dataset have been split into three different ratio of training and testing data which are 70%-30%, 80%-20% and 90%-10%. The aim of using different types of ratio is to identify the most suitable ratio that gives the best performance for the classifier model. The ratio 80%-20% is the most popular set used by the author such as Irmand & Astriratma (2020), Alawi et al., (2024) and Pal & Patel (2020) which perform average above 70% in terms of accuracy. The ratio 70%-30% is used by Yang et al. (2024) obtained result of accuracy by 0.743, precision is 0.739, recall 0.737 and F1-Score is 0.736 using the SVM algorithm. The 90%-10% ratio is

barely used in the poetry classification system, however it achieved close to perfect score on testing data as it was train and feed with more training data compared to other ratio.

3.4 Design

Table 3.9 shows the design phase which include the design of system flowchart, GUI, pseudocode and system architecture.

Table 3.9 Design Phase

Objectives	Phase	Activity	Method/Tools	Expected Deliverables
To develop a prototype of Malay *Pantun* theme classifier using SVM algorithm.	Design	**Activity :** Design **Task :** • Design the flowchart. • Design the system architecture. • Design the pseudocode. • Design the GUI.	Tool: Draw.io, Canva.	1.System flowchart. 2.System Architecture. 3.Malay *Pantun* theme classification pseudocode. 4.Template of Malay *Pantun* theme classification prototype GUI.

3.4.1 System Flowchart

The Malay *Pantun* theme classification process began with input of the Malay *Pantun* from the user. The input forwarded to the SVM classifier to identify and undergo preprocessing to ensure the input met the model classifier requirements. Later feature extraction is followed. The word is converted into numerical form. The model classified based on the past trained data and the output including types of theme, model performance and other related *Pantun* is shown on the GUI. Figure 3.3 shows the system flowchart of Malay *Pantun* theme classification prototype.

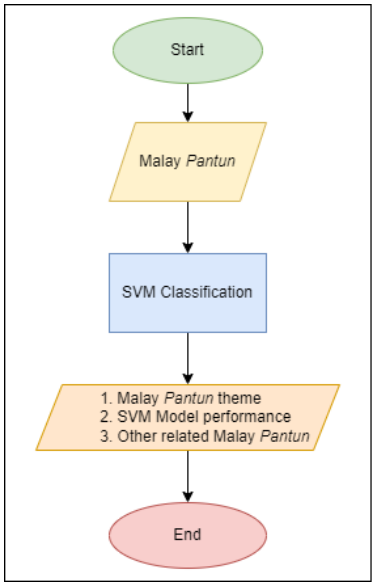


Figure 3.3 Flowchart of Pantun Theme Classification System

3.4.2 GUI Design

A GUI give a simple and easy to access component for the user to use the Malay *Pantun* theme classifier. Malay *Pantun* theme classifier comes with GUI that allows the user to insert the Malay *Pantun*. The model preprocess the input by user and the input is classified by the SVM to determine the types of the *Pantun*. The result showed the *Pantun* type, model performance and related *Pantun*. Figure 3.4 shows the Malay *Pantun* theme classifier GUI.

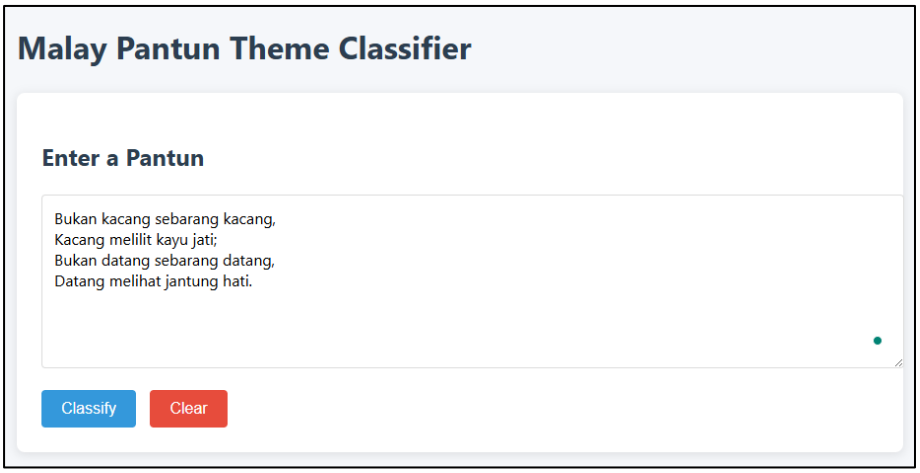


Figure 3.4 Malay Pantun Theme Classifier GUI

3.4.3 Pseudocode

The classifier begun input of the *Pantun* dataset. The dataset were processed through preprocessing techniques. Text segmentation is when the stanza document, d_i, is separated into two different part which is the indicator and meaning part is labelled as, S_i. Then, case folding is applied at S_i, to change all character's case to lower case, S_2. S_2 is broken into word which called as tokenization, F_j. Then each word that overlap with the stopword were removed as it has no significant impact on the *Pantun* meaning. Then, this words is converted to its root word which the step is called stemming.

Processed data undergo featured extraction using the method TF-IDF. TF-IDF vectorize this stemmed words and rank it based on the number of words in each stanza document and also in the whole document. When the feature is extracted, the data split into training and testing dataset. The training dataset was used for training the model while the testing dataset was used after the model to assess its performance based on the key metrics. Figure 3.5 shows the pseudocode of Malay *Pantun* theme classifier.

```
Begin
        INPUT: d1,d2,….,di number of the stanza document where Each di contain Number of
        Pantun. (Cn, the number of themes)
        For each themes Cn Do, For each stanza document dj for Cn Do
        Separate di into two different part indicator and meaning part, Si.
        Case folding the Si to lowercase the text S2.
        Tokenize the S2 into features S2→Fj
        Delete Stop word, number, the special character from Fj→F1j
        Stemming the F1j →F2j
        Count the frequency for all F2j
        Set the T1 threshold for term frequency and T2 for Pantun frequency
        Select the F2j which frequent >= T1
        Count the Pantun frequent >= T2
        Term Weight =Term frequent/ No. Pantun frequent
        Apply TF-IDF between the di
        Data Splitting
        Train the model using SVM
        Evaluate the model
        Test the model
        Classify a Pantun
End
```

Figure 3.5 Pseudocode of Malay Pantun Theme Classifier

3.4.4 System Architecture

System architecture is the design of system structure, by outlining the components, and interactions between components. It served as the blueprint in the beginning of the prototype development in order to ensure the development process is running smoothly on the track. The Malay *Pantun* dataset undergo preprocessing phase to ensure data is consistent, reliable and error-free. It is necessary to deal with the data missing values or inconsistencies of the data.

The processed data followed with the feature extraction process, using TF-IDF in order to vectorize each token by calculating the number of the word appeared in the text document and in the whole document. The main idea of feature extraction is to reduce the feature vector after text processing as it reached ten thousand even hundreds of thousands. Therefore, the feature selection is required to reduce the feature dimension and enhance classification efficiency (Zhu et al., 2023). SVM used the labelled training dataset to learns

and classify *Pantun* correctly. Figure 3.6 shows the system architecture of the Malay *Pantun* theme classification prototype.

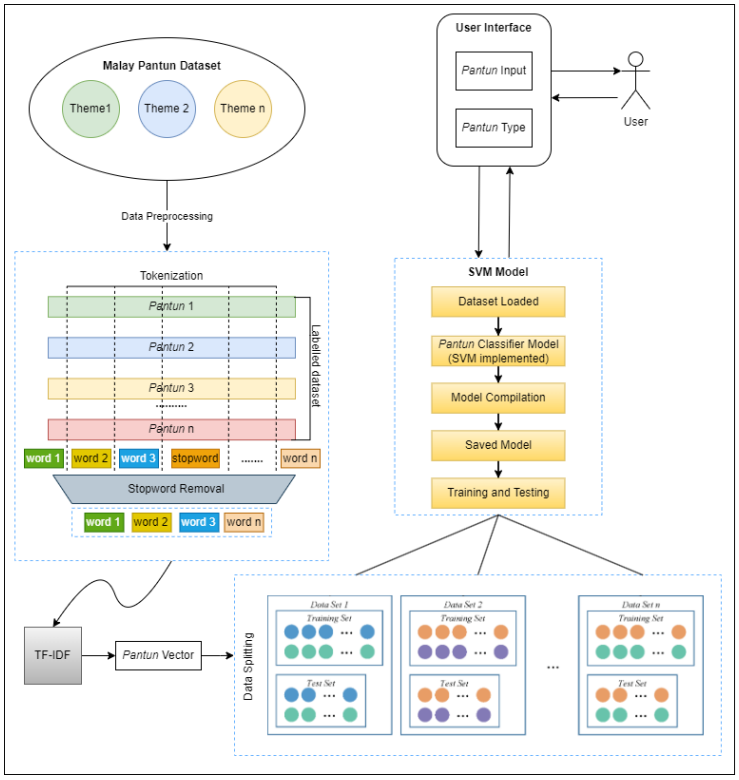


Figure 3.6 System Architecture of Malay Pantun Theme Classifier

(*Source*: Zhu et al., 2023; Yang et al., 2024)

3.5 Implementation

Table 3.10 shows the implementation phase which include implementation of SVM algorithm and the model training.

Table 3.10 Implementation Phase

Objectives	Phase	Activity	Method/ Tools	Expected Deliverables
To develop a prototype of Malay *Pantun* theme classifier using SVM algorithm	Implementation	**Activity :** Implementation **Task :** • Create a prototype using the designed prototype • Implement the SVM algorithm using Python. • Train the model using training dataset.	Tool : Python	A well-trained model that can classify the Malay *Pantun* into its distinct types using SVM classifier ready for user.

3.5.1 Model Development

Next, the model is developed based on the system architecture and flowchart, GUI design, and the pseudocode. The training dataset used to train the model. The SVM classifier trained continuously with different parameters and data ratios. After the training session is completed, the model is saved to reduce the number of training.

3.6 Evaluation

Table 3.11 shows the evaluation phase

Table 3.11 Evaluation Phase

Objectives	Phase	Activity	Method/Tools	Expected Deliverables
To evaluate the performance of the SVM algorithm for Malay *Pantun* theme classifier.	Evaluation	**Activity :** Evaluate prototype **Task :** • Perform test to assess the model performance. • Compare the performance metrics using different parameters.	Tool: Python Method: • Performance metrics: such as accuracy, precision, recall, and F1-score • Cross-validation	Result of model performance based on: 1.Accuracy 2.Precision 3.Recall 4.F1-Score

3.6.1 Model Testing

The saved model are available for the user to explore and test the model using testing dataset. Testing dataset is the data that are unlabelled. Model testing is crucial to determine either the model is well-trained, over trained or otherwise. The model is well-trained as it generated the best performance and classified the pantun into its corresponding theme correctly. Over-trained model is unnecessary as the model only memorized the *Pantun* and its labelled theme which result in the testing dataset to be misclassified, or the performance achieved is unsatisfactory (Reitermanova, 2010).

3.6.2 Evaluation Metrics

Evaluation is essential as the system performance is measured. There are a few measurement evaluations metric that are used to measure the model performance (Alawi et al., 2024; Irmanda & Astriratma, 2020; Noh & Basri, 2021; Yang et al., 2024). The evaluation metrics used are accuracy, precision recall, and F1-score, and confusion matrix used to visualize model performance.

a) Accuracy

Accuracy is the most used evaluation metric for text classification tasks. The metric were simple and interpret the measurement of the prediction is correct. True Positive (TP) is number of word in *Pantun* stanza document that assigned correctly to its themes. True Negative (TN) is number of word in *Pantun* stanza document assigned incorrectly to its themes. False Positive (FP), which is the number of word in *Pantun* stanza document incorrectly assigned to the themes and False Negative (FN) is the number of word in *Pantun* stanza document correctly rejected assigned to this category. The equation of accuracy is as shown in (3).

$$Accuracy = \frac{TP+TN}{TP+FP+FN+TN} \times 100\% \quad (3)$$

b) Precision and Recall

Precision is the number of predicted *Pantun* predicted correctly by the classifier, while recall is number of *Pantun* predicted through the whole *Pantun* document. It can be defined as shown in the (4) and (5)

$$Precision = \frac{TP}{TP+FP} \times 100\% \quad (4)$$

$$Recall = \frac{TP}{TP+FN} \times 100\% \quad (5)$$

c) F1 Score

F1-score is the combination of precision and recall together. This combination is also known as the mean of precision and recall.

$$F1 - Score = \frac{2PR}{2TP+FN+FP} \quad (6)$$

d) Confusion Matrix

After the result of the Malay *Pantun* metric evaluation is retrieved, a comparison is made in order to determine the best efficiency achieved by the model. Table 3.12 shows confusion matrix 2x2 is made up of actual result and predicted results.

Table 3.12 Confusion Matrix

Actual	Predicted	
	Positive	Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

e) Cross-validation

Cross validation is the methods that used for the data resampling techniques such as the stratified sampling. This technique was suitable when the data have multi-class and each of the class has even and uniform number of records. Biased occurred and only pick the data of one class overshadowed the other class. As an example, Malay *Pantun* contains eleven different themes, and six of the themes were selected for the research. Each themes contains uniform number of *Pantun*, so the best methods for the data resampling or splitting is the stratified sampling. Cross-validation allows the model to be implemented with other algorithm and parameters to compare the results with the developed model (Berrar, 2019).

3.7 Gantt Chart

Figure 3.7 shows the Gantt chart which serve as the project milestones.

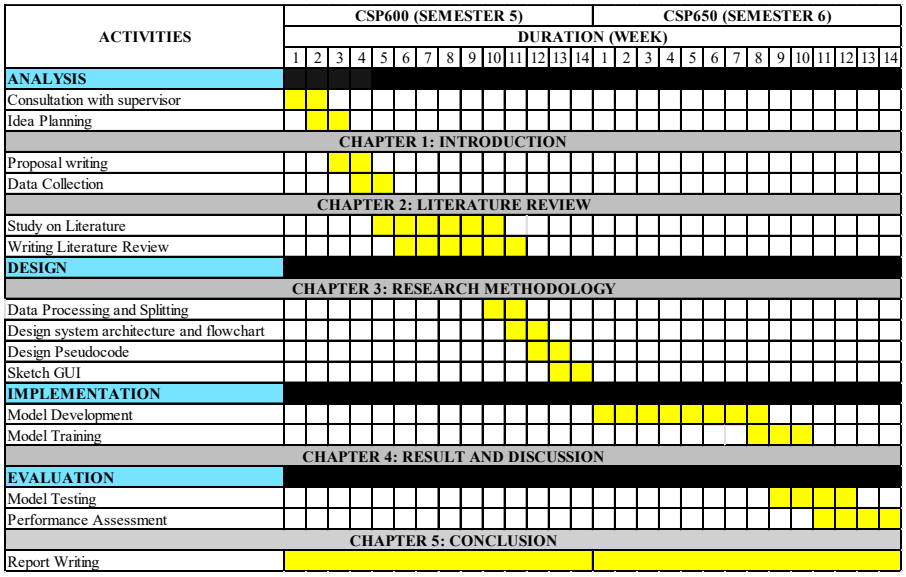


Figure 3.7 Gantt Chart

3.8 Conclusion

In conclusion, this chapter discussed about the research methodology alongside with the phases involve such as the preliminary study, design, implementation and evaluation phase. Preliminary study is focused on research finding for the articles and journal through online databases. Malay *Pantun* dataset were collected from the book *Kumpulan Pantun Melayu* Bakar (2018). The collected dataset is processed to ensure reliability and efficiency. The design phase involved the design of system architecture, flowchart, and pseudocode of the Malay *Pantun* theme classifier. The GUI was developed using Python. Then, SVM classifier were implemented into the prototype. In this phase, the model is trained using different parameters to achieved the best performance to the model. Finally, the evaluation phase, assess the model performance using key metrics such as accuracy, precision, recall, and F1-score using the confusion matrix.

CHAPTER 4

RESULT AND FINDING

This chapter offers a comprehensive overview of the prototype's output through user interface of the web-based system. The prototype was developed using Python, and several frameworks and libraries were incorporated to ensure functionality and performance. The web interface is powered by the Flask framework, and evaluation metrics are computed, and visualisations such as charts and performance summaries are produced using Scikit-learn (sklearn). The Support Vector Machine (SVM) algorithm that was manually created, hardcoded, and optimised to maximise model performance.

4.1 Results for Objective 1

Regarding to objective one of the project, which is to identify the implementation of SVM algorithm in text and poetry classification and its features., a comprehensive reading and analysis in two major aspects has been conducted. The first aspect goes into a detailed analysis of the relevant and related research on the SVM algorithm. The main purpose of this research is to unfold the use of SVM algorithms in the poetry classification. The project successfully obtained a strong foundation of comprehensive understanding on the implementation of the SVM algorithm used in text classification related to poetry and its features. The second aspect is focused on processed the data. This processing involve text segmentation, text cleaning, case folding, tokenization, stopword removal, stemming and features extraction. These techniques gave a significant impact on improving the quality of the dataset and make it more reliable and suitable for the SVM model training. By performing a deeper examination of these theoretical knowledge in the literature is important as well as the practical factors involving the data preprocessing.

4.1.1 Analysis of Literature Review on SVM Algorithm

Kernel selection is important, as it significantly affects SVM's capability for classification. Numerous studies that demonstrate experiments with linear, polynomial, and radial basis function (RBF) kernels are included in the review. This shows the flexibility of SVM for fitting data structures that are linearly separable as well as non-linear. Kernel tuning is important in achieving the best classification performance. The outcome is directly consistent with the experimental design of this project. The literature reviewed also suggest TF-IDF-based feature extraction as a dominant input representation strategy when working with textual or poetic data. Frequency-weighted vector models can effectively represent the linguistic features needed to classify subtle thematic distinctions, such as those in *Pantun*.

Precision, recall, and F1-score are reported occasionally, but accuracy is the main key metrics for performance evaluation. Accuracy provided a general overview of model success. This project has bridged the research in Malay literary field created from earlier work which lack of in-depth metric analysis by conducting a more comprehensive multi-metric evaluation of the SVM classifier. SVM's continued widespread use with traditional vector-based features attests to the fact that it is a baseline model that is dependable and understandable rather than context-sensitive. The literature's acknowledgement of this limitation and the potential for future developments suggest for hybrid approaches for future research in this study.

Furthermore, the literature review shows that SVM consistently performs competitively across a variety of poem classification tasks, even when compared to other algorithms like Naïve Bayes or Logistic Regression. Even though SVM is not always the best model, it is often commended for its ability to generalise and perform well with smaller datasets. The 720 labelled *Pantun* dataset used in this study closely resembles these characteristics. Finally, hyperparameter

tuning such as selecting C, gamma, and iteration are among the common problems with SVM that are listed in the literature. This study employs manual model design, methodical parameter configuration and few experiments were conducted for the testing and evaluating in order to find the best parameters for the SVM model to give the best possible results. In conclusion, the review of the literature provides a solid foundation for using SVM for *Pantun* theme classification tasks. It highlights standard procedures, recognises the flexibility and efficiency of the algorithm, and directs the methodological choices made for this project.

4.1.2 Dataset Pre-processing

To ensure that the SVM classifier identified the correct themes of Malay *Pantun*, dataset preprocessing steps were carried out on the raw dataset. These processes is to transform the original *Pantun* text into a clean, standardised, machine-readable format suitable for feature extraction and classification. The dataset, which comprised 720 *Pantun* entries extracted from "*Koleksi Pantun Melayu*" book, which being segmentized, cleaned, case folded, tokenized, stopword removed, and stemmed. These were done with Python using custom functions and appropriate regular expressions. The stage were recognised in NLP practices, particularly for morphologically complex language structure like Malay languages. These procedures significantly enhanced the classifier's ability to recognise thematic and semantic patterns in the *Pantun* stanzas while also ensuring consistency across the dataset.

4.1.2.1 Text Segmentation

A segmentation function was using re.split() function, which enables string splitting according to regular expression patterns. A specific delimiters like commas, semicolons, and exclamation points ([;,!?]), followed optionally by whitespace, was the function's task in this instance. In accordance with standard

Pantun formatting, this ensures a clean line division. After segmentation, a list comprehension ([line.strip() for line in re.split()]) removed any leading and trailing whitespaces and omitted empty segments. Figure 4.1 show the code for *Pantun* segmentation.

```
def segment_pantun(self, text):
    """Segment pantun into meaningful lines"""
    if not isinstance(text, str):
        return ""

    # Split on common Malay line-ending punctuation including commas
    lines = [line.strip() for line in re.split(r'[;,!?]\s*', text) if line.strip()]

    if len(lines) >= 4:
        return ' '.join(lines[2:4])  # 3rd and 4th lines
    return ' '.join(lines)  # fallback
```

Figure 4.1 Code for Pantun Segmentation

To preserve the meaningful part of the *Pantun*, a conditional check was used to ensure that there were four or more split segments. If this condition was met, the last two lines, which typically contained the core message, were extracted using the slicing operation lines [2:4]. These two lines were then joined back into a single, coherent sentence using ' '.join(lines[2:4]). In cases where fewer than four segments were found, a fallback operation was employed to join all available lines to ensure that no input was entirely discarded.

4.1.2.2 Text Cleaning

The extracted lines were cleaned to remove extraneous characters and irregularities following segmentation. The cleaning function employed re.sub() operations to eliminate symbols. By eliminating all characters that were not word characters, whitespace, or basic punctuation, the first regular expression, re.sub(r'[^\w\s;,!?]', '', text), preserved linguistic integrity while removing artefacts like emojis, HTML tags, and non-Malay special characters. This ensured that the corpus was sanitised and ready for examination. Figure 4.2 shows the code for text cleaning.

```python
def clean_text(self, text):
    """Basic text cleaning"""
    if not isinstance(text, str):
        return ""

    # Remove special characters except Malay letters and basic punctuation
    text = re.sub(r'[^\w\s;,!?]', '', text)
    # Normalize whitespace
    text = re.sub(r'\s+', ' ', text).strip()
    return text
```

Figure 4.2 Code for Text Cleaning

A different re.sub() pattern that merged several spaces into one space (re.sub(r'\s+',", text)) was used to normalise excessive whitespace. Then, using .strip(), any leading or trailing whitespace was removed. The formatting became uniform across the dataset as a result of this modification. The outcome was a corpus of regularly arranged text lines that were syntactically correct and semantically relevant for further language processing.

4.1.2.3 Case Folding

The case folding step converted all of the capital letters in the cleaned text to lowercase using Python's .lower() method. This step was essential to textual normalisation because it allowed the model to treat "Yang" and "yang" as the same word. By standardising the character casing and preventing the duplication of features that differed only in case, this step reduced dimensionality and improved pattern recognition. Figure 4.3 shows the code for case folding the *Pantun* text.

```python
def case_fold(self, text):
    """Convert text to lowercase"""
    if not isinstance(text, str):
        return ""
    return text.lower()
```

Figure 4.3 Code for Case Folding

4.1.2.4 Tokenization

After normalisation, the text was tokenized where it is divided into distinct word units or tokens using a regular expression pattern. In order to record word boundaries with letters and apostrophes, the function re.findall(r'\b[\w\']+\b', text) were used. Given that apostrophes are commonly used to indicate contractions and possessive forms in Malay texts, this was especially appropriate. Figure 4.4 shows the code for tokenization

```
def tokenize(self, text):
    """Tokenize text into words"""
    if not isinstance(text, str):
        return []
    # Malay-specific tokenization that keeps common contractions
    return re.findall(r'\b[\w\']+\b', text)
```

Figure 4.4 Code for Tokenization

The end result was a list of tokens for each word in a *Pantun*. For instance, the phrase "yang baik budi yang indah bahasa" can be broken down into the words ["yang", "baik", "budi", "yang", "indah", "bahasa"] This tokenization improved the linguistic sensitivity and performance of the model, allowing the subsequent pre-processing operations, such as stemming and stopword removal, to operate at a finer level.

4.1.2.5 Stopword Removal

Stopword removal was used to eliminate high-frequency, low-value words that contribute less significant to thematic understanding. The carefully chosen list of Malay stopwords included over a hundred words that are commonly used in literary and conversational contexts. Every token from the tokenised list was compared using this stopword list. Figure 4.5 shows the code for stopword removal.

```
def remove_stopwords(self, tokens):
    """Remove common Malay stopwords"""
    if not isinstance(tokens, list):
        return []
    malay_stopwords = {…
    }
    return [token for token in tokens if token not in malay_stopwords]
```

Figure 4.5 Code for Stopword Removal

4.1.2.6 Stemming

The stemming reduced derived words to their most basic forms. This was done using the stem() function of the Sastrawi stemmer, which is designed for the Malay language. To pass each token through the stemmer, a list comprehension was employed. The operation self.stemmer.stem(token) processed each word by identifying its morphological structure and removing common affixes such as prefixes (such as "ber-," "men-," and "ter-") and suffixes (such as "-kan" and "-i"). For instance, "berjalan" were stemmed to "jalan," and "memahami" were reduced to "faham." This transformation significantly improved the consistency of the dataset by ensuring that different forms of the same base word were recognised as a single entity. Figure 4.6 shows the code for word stemming.

```
def stem(self, tokens):
    """Stem Malay words using Sastrawi"""
    if not isinstance(tokens, list):
        return []
    return [self.stemmer.stem(token) for token in tokens]
```

Figure 4.6 Code for Word Stemming

The project's initial objective was achieved, according to a thorough examination of identifying the implementation of SVM algorithm in text and poetry classification and its features. This goal was to investigate how SVM works as a supervised learning model when working with structured poetic data, specifically when it comes to classifying traditional Malay *Pantun*. The literature review provided a critical foundation by looking at previous studies

that used SVM and other machine learning methods in the textual and poetic domains.

4.2 Results for Objective 2

This section displays the outcomes of the second project goal, which was to develop a prototype of Malay Pantun theme classifier using SVM. The results are divided into three major sections. The first section discusses how to prepare and represent the processed input dataset. The second section explains the implementation of the custom SVM classifier algorithm. The third section showcases the interface prototype, highlighting its usability, prediction output, and user experience.

4.2.1 Input Representation

Following the completion of all pre-processing procedures, such as segmentation, cleaning, case folding, tokenization, stopword removal, and stemming, the refined dataset was prepared for the classification stage. This dataset, which was numerically represented using the Term Frequency-Inverse Document Frequency (TF-IDF) technique. This processed dataset was used as input for the SVM classification model. The linguistic features of the *Pantun* maintaining semantic relevance, improving the classifier's ability to distinguish between thematic nuances. Figure 4.7 shows the output for Malay *Pantun* final text which is the input used for classifying.

PANTUN	TEMA	final_text
Yang kurik kundi, yang merah saga; Yang baik budi, yang indah ba	PANTUN BUDI	baik budi indah bahasa

Figure 4.7 Output of the Final Text

The dataset was intentionally split into three configurations for the training and testing processes: 90:10, 80:20, and 70:30 ratios, respectively. A multi-ratio approach was employed to fully assess the model's learning and generalisation abilities across a range of data sizes. In each instance, the larger subset was used to train the model, while the smaller portion was retained only for testing. The input representation, built on top of TF-IDF and based on clean, semantically consistent tokens, proved to be a solid foundation for the SVM classifier's subsequent learning phase. This representation significantly enhanced the system's ability to identify and categorise Malay *Pantun* themes based on their root-level lexical patterns.

4.2.2 Implementation of SVM Algorithm

This section discusses in detail the specific implementation features of the Support Vector Machine (SVM) algorithm for classifying Malay *Pantun* themes. Efficiency, interpretability, and modularity are given top priority in this manually constructed Python model. The code integrates essential features such as kernel selection, alpha optimisation, multi-class support, model saving/loading, and accurate prediction to efficiently handle the classification of poetic texts.

The core component, the CustomSVM class, inherits from BaseEstimator and ClassifierMixin to make it compatible with the Scikit-learn ecosystem. The initialisation method enables users to customise the model's flexibility and convergence behaviour to their own data by defining key hyperparameters such as kernel, C, gamma, degree, max_iter, and tol. Figure 4.8 shows the code for CustomSVM.

```python
class CustomSVM(BaseEstimator, ClassifierMixin):
    def __init__(self, kernel='rbf', C=1.0, gamma=0.1, degree=3, max_iter=1000, tol=1e-4):
        self.kernel = kernel
        self.C = C
        self.gamma = gamma
        self.degree = degree
        self.max_iter = max_iter
        self.tol = tol
        self.models = {}
        self.label_to_int = {}
        self.int_to_label = {}
```

Figure 4.8 Code for CustomSVM Constructor

The kernel function is a crucial component that dictates how input data is transformed into a higher-dimensional feature space. The RBF kernel, which applies a Gaussian similarity metric regulated by the gamma parameter, linear (dot product), and polynomial (using adjustable degree) are the three types that the _kernel_function() method handles as shown in the Figure 4.9. These kernels allow the classifier to adapt to both linearly and non-linearly separable data while used to explore and produced various model performance.

```python
def _kernel_function(self, x, y):
    if self.kernel == 'linear':
        return np.dot(x, y)
    elif self.kernel == 'rbf':
        return np.exp(-self.gamma * np.linalg.norm(x - y) ** 2)
    elif self.kernel == 'poly':
        return (np.dot(x, y) + 1) ** self.degree
    else:
        raise ValueError("Unsupported kernel type")
```

Figure 4.9 Code for Kernel Function

The _compute_kernel_matrix() method is then used to create a symmetric kernel matrix, where each element is computed using the designated kernel function. Because it computes inner products directly, eliminating the need to explicitly transform data into high-dimensional space, this matrix is essential to SVM operations. Figure 4.10 shows the code for _compute_kernel_matrix() function.

```python
def _compute_kernel_matrix(self, X):
    n_samples = X.shape[0]
    K = np.zeros((n_samples, n_samples))
    for i in range(n_samples):
        for j in range(n_samples):
            K[i, j] = self._kernel_function(X[i], X[j])
    return K
```

Figure 4.10 Code for Computing Kernel Matrix

Training is done by the _train_binary_svm() method, which uses a one-vs-rest strategy to support multi-class classification. For each class, a binary classifier is trained against all other classes. During training, the method iterates through the dataset, modifying Lagrange multipliers (alpha values) in compliance with the SVM margin violation requirements. The convergence is monitored using the L2-norm difference between successive alpha vectors. A bias term b is calculated by averaging over valid support vectors, as is shown at the end of the process. Figure 4.11 shows the code for _train_binary_svm().

```python
def _train_binary_svm(self, X, y):
    if hasattr(X, "toarray"):
        X = X.toarray()

    X = np.array(X)
    y = np.array(y)

    # Ensure X is 2D
    if X.ndim == 1:
        X = X.reshape(-1, 1)

    n_samples = X.shape[0]
    alpha = np.zeros(n_samples)
    b = 0.0
    K = self._compute_kernel_matrix(X)

    pbar = tqdm(total=self.max_iter, desc="Training Binary SVM", ncols=80)
    for epoch in range(self.max_iter):
        alpha_prev = np.copy(alpha)
        for i in range(n_samples):
            output = np.sum(alpha * y * K[:, i]) + b
            error = output - y[i]

            if (y[i] * error < 1 and alpha[i] < self.C) or \
               (y[i] * error > 1 and alpha[i] > 0):
                alpha[i] -= y[i] * error / (K[i, i] + 1e-8)
                alpha[i] = np.clip(alpha[i], 0, self.C)

        diff = np.linalg.norm(alpha - alpha_prev)
        pbar.set_postfix({'diff': f'{diff:.5f}'})
        pbar.update(1)
    pbar.close()

    # Compute bias
    support_idx = np.where((alpha > 1e-5) & (alpha < self.C))[0]
    if len(support_idx) > 0:
        b_sum = 0
        for i in support_idx:
            b_i = y[i] - np.sum(alpha * y * K[:, i])
            b_sum += b_i
        b = b_sum / len(support_idx)

    return alpha, b, X, y
```

Figure 4.11 Code for Binary Train SVM

The major orchestration occurs in the fit() method. In this case, all unique class labels are mapped to integers, and each class is trained as a binary classification. Each class-specific model stores learnt parameters like alpha, labels, bias, and support vectors. By generating a dictionary of models, each of which represents a single class in the multi-class problem, the process satisfies the one-vs-rest approach. Figure 4.12 shows the code for fit() function.

```
def fit(self, X, y):
    if hasattr(X, "toarray"):
        X = X.toarray()
    X = np.array(X)
    y = np.array(y)

    if X.ndim == 1:
        X = X.reshape(-1, 1)

    unique_labels = sorted(set(y))
    self.label_to_int = {label: i for i, label in enumerate(unique_labels)}
    self.int_to_label = {i: label for label, i in self.label_to_int.items()}

    y_encoded = np.array([self.label_to_int[label] for label in y])

    for class_idx, class_label in self.int_to_label.items():
        print(f"🚀 Training for class '{class_label}' vs rest")
        binary_y = np.where(y_encoded == class_idx, 1, -1)
        alpha, b, X_sv, y_sv = self._train_binary_svm(X, binary_y)
        self.models[class_label] = {
            'alpha': alpha,
            'b': b,
            'X': X_sv,
            'y': binary_y
        }

    # Needed for app.py compatibility
    self.classes_ = np.array(list(self.models.keys()))
```

Figure 4.12 Code for Training SVM

This SVM model uses a custom preprocessor that includes TF-IDF transformation to preprocess the *Pantun* dataset. The processed features X and related labels y are then separated into training and testing subsets using stratified sampling. The 'poly' kernel is instantiated, and then the TF-IDF vectors are used to train the CustomSVM classifier. Training iterations and progress are displayed using a tqdm progress bar since the project is absence of epochs as the progress indicator. Figure 4.13 displays the train_model.py model training flow.

```
start_train = time.time()
classifier = CustomSVM(kernel='poly', C=1.0, gamma=0.1, max_iter=100)

# Simulate progress bar
classifier.fit(X_train.toarray(), y_train)
time_train = time.time() - start_train
print(f"✅ Model trained in {time_train:.2f} seconds")
```

Figure 4.13 Code for Invoke CustomSVM

The model's performance is then evaluated by the script using a custom EvaluationMetrics class. It computes accuracy, precision, recall, and F1-score and generates a confusion matrix. The confusion matrix is saved as an image for visualisation, and the overall evaluation metrics are saved in a JSON file. Figure 4.14 shows the code for evaluation metrics.

```
y_pred = classifier.predict(X_test.toarray())
evaluator = EvaluationMetrics(y_test.values, y_pred, classes=np.unique(y_test))
metrics = evaluator.calculate_metrics()

#confusion metrics
print("\n📉 Confusion Matrix:")
cm = evaluator.get_confusion_matrix()
print(cm)

time_eval = time.time() - start_eval
with open('data/processed/eval_metrics.json', 'w') as f:
    json.dump(convert_numpy_types(metrics), f, indent=4)

print("\n🔍 Overall Performance:")
print(f"- Accuracy : {metrics['accuracy']:.4f}")
print(f"- Precision: {metrics['precision']:.4f}")
print(f"- Recall   : {metrics['recall']:.4f}")
print(f"- F1-Score : {metrics['f1']:.4f}")

print("\n📊 Per-Class Report:")
report_df = pd.DataFrame(metrics['class_report']).transpose()
print(report_df.round(3))
```

Figure 4.14 Code for Invoke Evaluation Metrics

The trained model can be serialised using Joblib and saved to disc using the save_model() method reduce the need for retraining. A matching load_model() static method ensures that saved classifiers are effectively restored during deployment. Figure 4.15 shows the code for the model saving function.

```python
def save_model(self, filepath):
    os.makedirs(os.path.dirname(filepath), exist_ok=True)
    joblib.dump(self, filepath)
    print(f"✅ Model saved to {filepath}")
```

Figure 4.15 Code for Model Saving

The _project() method determines decision which class has the highest score values by summing up all of the support vectors, each of which is weighted by its alpha value and label. Softmax normalisation is used by the predict_proba() function to generate probabilistic predictions. Figure 4.16 displays the prediction mechanism

```python
def _project(self, x, model):
    alpha = model['alpha']
    b = model['b']
    X_sv = model['X']
    y_sv = model['y']

    s = 0
    for i in range(len(X_sv)):
        s += alpha[i] * y_sv[i] * self._kernel_function(X_sv[i], x)
    return s + b

def predict(self, X):
    if hasattr(X, "toarray"):
        X = X.toarray()
    X = np.atleast_2d(X)

    all_scores = []

    for label, model in self.models.items():
        scores = np.array([self._project(x, model) for x in X])
        all_scores.append(scores)

    all_scores = np.array(all_scores)
    best_class_indices = np.argmax(all_scores, axis=0)
    predicted_labels = [list(self.models.keys())[i] for i in best_class_indices]

    return np.array(predicted_labels)

def predict_proba(self, X):
    if hasattr(X, "toarray"):
        X = X.toarray()
    X = np.atleast_2d(X)

    all_scores = []
    for label, model in self.models.items():
        scores = np.array([self._project(x, model) for x in X])
        all_scores.append(scores)

    all_scores = np.array(all_scores).T

    exp_scores = np.exp(all_scores - np.max(all_scores, axis=1, keepdims=True))
    probs = exp_scores / np.sum(exp_scores, axis=1, keepdims=True)
    return probs
```

Figure 4.16 Code for Prediction and Probability in User Interface

In conclusion, a comprehensive and trustworthy foundation for Malay *Pantun* theme classification prototype is provided by the code structure of the SVM algorithm. The functions included are evaluation metrics, multi-class handling, kernelized decision boundaries, and vectorised TF-IDF input. This section effectively replicates the whole machine learning pipeline, from preprocessing to model training, evaluation, and storage.

4.2.3 Prototype Interface

This section provides a detailed explanation of the features and development of the graphical user interface (GUI) made for the Malay *Pantun* theme classification prototype. The interface was developed using Flask for server-side logic and HTML, CSS, and JavaScript for client-side interaction. With the provided prototype's interface, users can input *Pantun*, perform real-time theme classification, examine evaluation metrics, and visualise data insights. Figure 4.17 show users can type or paste a Malay *Pantun* into a central input form in the main layout. Upon clicking the "Classify" button, the input is routed to the backend route /classify, where it is pre-processed and then vectorised using a loaded TF-IDF model. This vectorised input is fed into a pre-trained CustomSVM classifier, which classify the *Pantun*'s theme and along with prediction probabilities and examples of related *Pantun*.

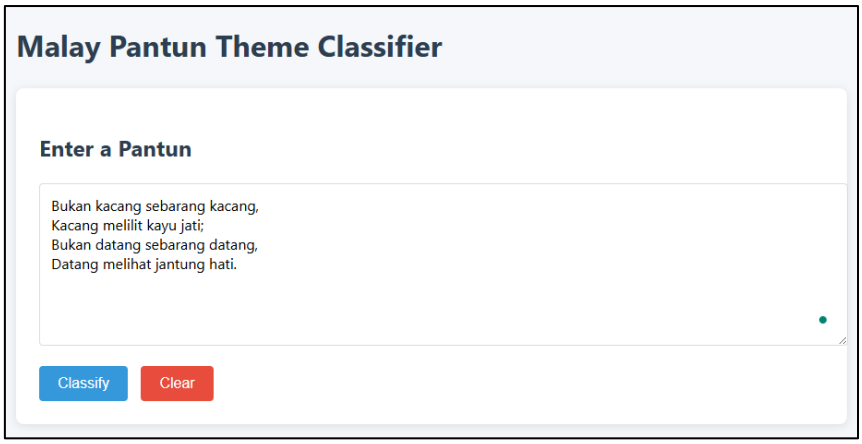


Figure 4.17 GUI of Malay Pantun Theme Classifier Prototype

A confidence bar visualisation, with each bar being clearly labelled and colour-coded, is also used to show the probability distribution across all available themes. This confidence bar show how confidence the model on classifying the input *Pantun* into their theme respectively. From the loaded dataset, three related *Pantun* examples that fit the classified theme are chosen at random and shown in a secondary panel. When the user enters a *Pantun* with a romantic theme, it successfully classifies it as "PANTUN PERCINTAAN" and

displays three related *Pantun* samples from the dataset, confirming the functionality and relevance of the system. Figure 4.18 shows, the GUI sample.

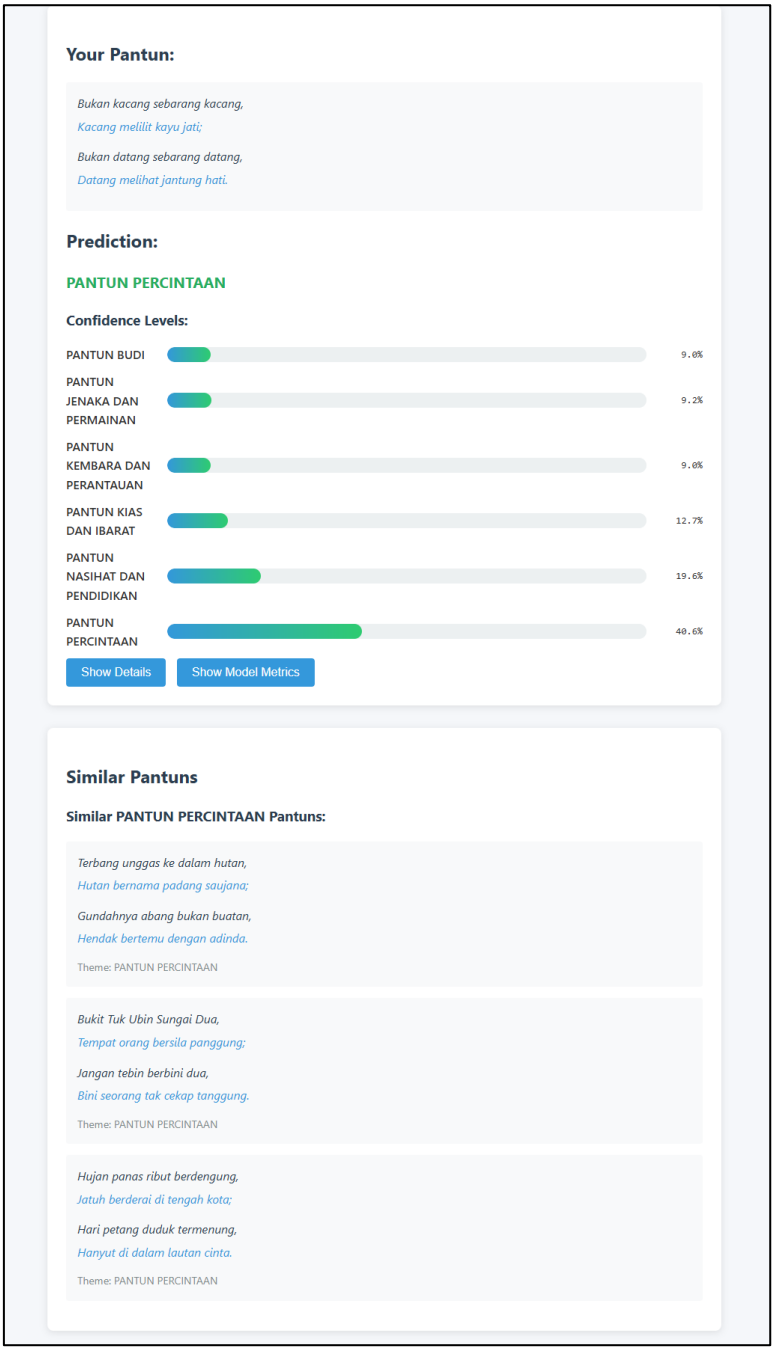


Figure 4.18 User Interface after Pantun Input

Another visualization of the model confidence level in classfying the Malay *Pantun* is visualized in the donut chart form. The donut chart can be accessed when the user clicked the show details button. Figure 4.19 shows the donut chart

visualization of the Malay *Pantun* distribution confidence level. When the mouse is hover on the donut chart it showed the percentage of the classification details.

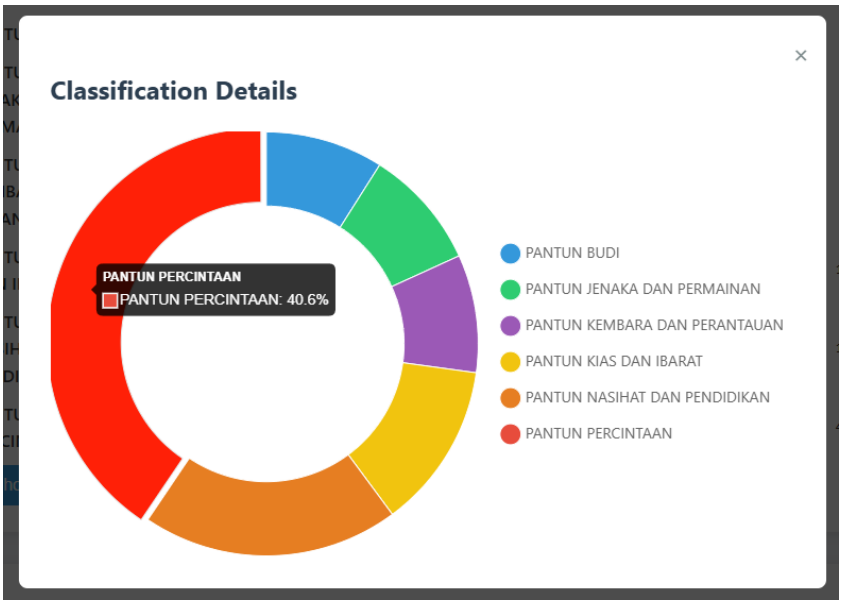


Figure 4.19 User Interface for Confidence Level Distribution – Donut Chart

The app.py implementation of the backend logic loads the required models and vectorizers first. This includes the specially trained SVM model as well as the serialised and data/models directory-stored TF-IDF vectorizer. To enable real-time classification, these saved resources are loaded at application startup. The primary classification functionality is located in the /classify route. Additional functionality is provided by a button called "Show Model Metrics," which sends an asynchronous request to the /evaluate route of the Flask server. After classifying a predefined test set, this method yields significant evaluation metrics like accuracy, precision, recall, and F1-score. The JavaScript logic dynamically generates a modal to show this data in a table format, as shown in Figure 4.20.

Model Evaluation Metrics

Evaluation Metrics (Split Ratio - 90/10)

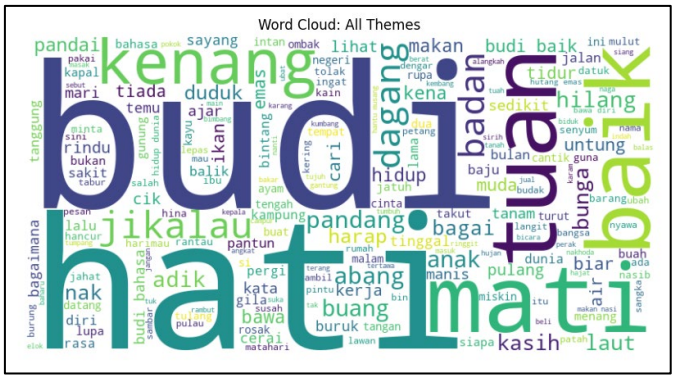
Metric	Value
Accuracy	0.667
Precision	0.664
Recall	0.667
F1-Score	0.664

Detailed Classification Report for Each Pantun Theme

Class	Precision	Recall	F1-Score	Support
PANTUN BUDI	0.846	0.917	0.880	12
PANTUN JENAKA DAN PERMAINAN	0.727	0.667	0.696	12
PANTUN KEMBARA DAN PERANTAUAN	0.692	0.750	0.720	12
PANTUN KIAS DAN IBARAT	0.600	0.500	0.545	12
PANTUN NASIHAT DAN PENDIDIKAN	0.500	0.500	0.500	12
PANTUN PERCINTAAN	0.615	0.667	0.640	12

Figure 4.20 User Interface for Model Evaluation Metrics

Users can access a comprehensive classification report by theme, which includes individual metric scores and support counts. This is further enhanced by visualisations such as a word cloud created with the WordCloud and Matplotlib libraries, and a theme distribution bar chart as shown in Figure 4.21 and Figure 4.22 respectively. Both of these data visualization shows the result from the collective 720 number of Malay *Pantun* that is gather 120 *Pantun* for each themes. By opening the metrics modal as an overlay, users can browse thorough performance insights without ever leaving the main classification interface. Clear labelling, feedback mechanisms, and consistent colour schemes ensure an intuitive and instructive user experience.


Word Cloud: All Themes
budi
hati
mati
tuan
baik
kenang
jikalau
dagang
pandai
makan
budi baik
badan
hilang
untung
bunga
bagaimana
tanggung
adik
abang
buang
kasih
laut
pandang
harap
anak
kerja
hidup
kena
lihat
sayang
bahasa
duduk
tiada
mari
rindu
sakit
bukan
nak
datang
diri
lupa
rasa
balik
ikan
temu
ajar
cari
bintang
emas
bawa
cerai
buruk
pergi
kata
gila
kampung
pantun
cik
lalu
bagai
tidur
sedikit
muda
baju
bulan
dunia
biar
pulang
manis
miskin
menang
air
buah
ada
nasib
sangka
tanam
turut
takut
tinggal
malam
rumah
siapa
lawan
matahari
rosak
susah
ambil
pintu
tangan
kain
rupa
negeri
tolak
ingat
jalan
datuk
senyum
nama
cantik
guna
barang
ubah
nyawa
bangsa
langit
bicara
perak
hujan
makan nasi
beli
itu
patah
balas
siang
mulut
ini
tuah
emas
hutang
alangkah
dengar
petang
kunang
tempat
kering
lama
jatuh
tengah
cinta
kepala
kayu
hidup dunia
gunung
minta
sini
pesan
hina
hancur
harsmau
rantau
buat
si
pangkat
terang
tak
bin
itu
pulau
tambut
tulang
elok
burung
jahat
bahasa
budi
sayang
ombak
ikan
ibu
ayam
tujah
dentum
kembang
tinggi
kuda
tak
bin
pulang
lawan
patah
balas
siang
nama
budak
jual
kurus
barang
ubah
nyawa
bicara
perak
hujan
makan nasi
beli
itu
sangka
nasib
ada
buah
air
menang
miskin
manis
malam
rumah
tinggal
takut
turut
tanam
langit
bangsa
petang
petang
kain
kain
tuah
emas
hutang
datuk
senyum
nama
cantik
guna

the confusion matrix. A functional test of the system prototype were assessed to ensure all component functionality. The model's performance were assessed using different data splitting ratios and kernel.

4.3.1 Evaluation of SVM Model

The assessment began by applying various data splitting ratios to the *Pantun* dataset, which contained 720 labelled entries: 70-30, 80-20, and 90-10 for training and testing. Three different kernel types which are linear, RBF, and polynomial (Poly) were used to test each configuration in order to figure the best classification performance. Of all the splits tested with the polynomial kernel, with the 90-10 ratio configuration had the highest accuracy (68.06%), followed by the 80-20 configuration (63.19%) and the 70-30 configuration (60.65%). These results clearly demonstrate how well the polynomial kernel captures the complex syntactic and semantic nuances of *Pantun* themes, which are often hard to distinguish due to overlapping contextual expressions and vocabulary. Table 4.1, the overall experiment result.

Table 4.1 Result Performance for SVM Model with Different Parameter

Total Dataset	Data Splitting				Accuracy		
	Training		Testing		Kernel		
	Percentage	Number of Dataset	Percentage	Number of Dataset	Linear	RBF	Poly
720 *Pantun*	90%	648	10%	72	0.6528	0.6667	0.6806
	80%	576	20%	144	0.5972	0.6319	0.6319
	70%	504	30%	216	0.5648	0.5926	0.6065

Even though it is expected that more training data (as in 90-10) would always result in better performance, the results show only a slight advantage over the 80-20 configuration. The polynomial kernel's better performance suggests that nonlinear decision boundaries are more appropriate for *Pantun* theme classification than the linear or RBF counterparts. The RBF kernel did not perform well in this scenario, due to insufficient parameter tuning, even though it was theoretically capable of representing complex patterns.

4.3.2 Model Performance

The model performance of the Malay *Pantun* theme classification prototype was evaluated using a confusion matrix, an essential diagnostic tool for evaluating the effectiveness of the SVM classifier. The model's performance in recognising each class in this multiclass classification scenario with six *Pantun* themes is detailed in the confusion matrix. The matrix offers data representation on both true positives and misclassifications. Figure 4.24 shows the confusion matrix.

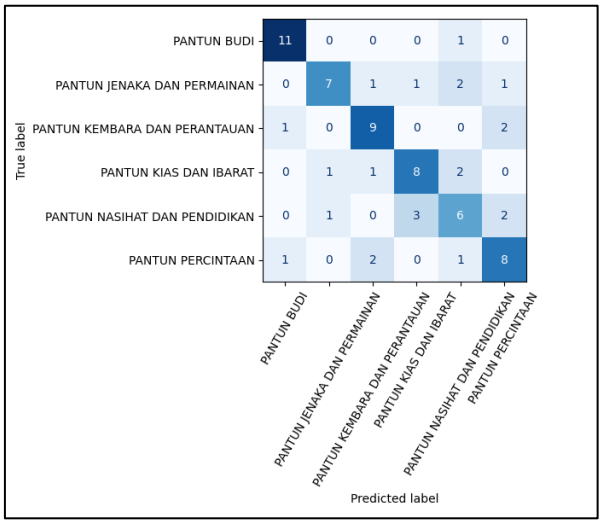


Figure 4.23 Confusion Matrix

This matrix provides thorough details regarding the model's ability to predict outcomes specific to a class. With only one sample being incorrectly classified, the theme "*Pantun* Budi" generated 11 accurate predictions, demonstrating the model's high confidence and discriminative power in identifying this category. However, themes like "*Pantun Jenaka dan Permainan*" and "*Pantun Nasihat dan Pendidikan*" exhibit higher levels of confusion with other classes, possibly as a result of overlapping lexical or semantic features. These overlapping features pose a greater challenge to the classifier, which operates with numerical representations of text.

This confusion matrix was used to compute evaluation metric The metric accuracy, which is the ratio of accurate predictions to total predictions, indicates that the classifier correctly predicted about 68.06% of all samples. The precision of the model, which measures the accuracy of positive predictions meaning that it correctly predicted a specific *Pantun* theme 68.30% of the time. The model's recall gauges how well it recognises all real instances of a class, shows that 68.06% of true theme instances were captured. As demonstrated by the F1 score of 0.6789, which integrates precision and recall into a single performance metric, the classification performance was modest but balanced. Table 4.2 shows the overall model performance.

Table 4.2 Overall Model Performance

Metrices	Result	Percentage
Accuracy	0.6806	68.06%
Precision	0.6830	68.30%
Recall	0.6806	68.06%
F1-Score	0.6789	67.89%

The per-class evaluation offers more specific details. "*Pantun Budi*" had the highest metrics of any category, with an F1 score of 0.880, a precision of 0.846, a recall of 0.917, and a thematic distinction. "*Pantun Kembara dan Perantauan*" and "*Pantun Kias dan Ibarat*" also showed balanced performance, with F1 scores of 0.720 and 0.667, respectively. "*Pantun Nasihat dan Pendidikan*" performed poorly with an F1 score of 0.500, suggesting frequent misclassification, due to the samples' lack of distinctive lexical or structural markers. These findings demonstrate that while the classifier struggles with less distinct thematic categories, it can distinguish between more distinct ones. To improve this performance, it might be necessary to use deeper models, more training data, or advanced text representation methods like word embeddings. By experimenting with ensemble approaches or modifying kernel parameters, accuracy was increased. Figure 4.25 shows the overall performance and result that have been achieved in the experiment.

```
📉 Confusion Matrix:
[[11  0  0  0  1  0]
 [ 0  7  1  1  2  1]
 [ 1  0  9  0  0  2]
 [ 0  1  1  8  2  0]
 [ 0  1  0  3  6  2]
 [ 1  0  2  0  1  8]]

🔍 Overall Performance:
- Accuracy : 0.6806
- Precision: 0.6830
- Recall   : 0.6806
- F1-Score : 0.6789

📊 Per-Class Report:
                              precision  recall     f1  support
PANTUN BUDI                       0.846   0.917  0.880     12.0
PANTUN JENAKA DAN PERMAINAN       0.778   0.583  0.667     12.0
PANTUN JENAKA DAN PERMAINAN       0.778   0.583  0.667     12.0
PANTUN KEMBARA DAN PERANTAUAN     0.692   0.750  0.720     12.0
PANTUN KIAS DAN IBARAT            0.667   0.667  0.667     12.0
PANTUN NASIHAT DAN PENDIDIKAN     0.500   0.500  0.500     12.0
PANTUN PERCINTAAN                 0.615   0.667  0.640     12.0
```

Figure 4.24 Output Display at Terminal

4.3.3 Functionality Testing

The goal of functionality testing was to evaluate the responsiveness and performance of all interactive components and backend features for the Malay *Pantun* theme classification prototype. The objective was to ensure smooth usability, stable core feature execution, and accurate prediction behaviour for end users. A total of six functionality test cases were developed and executed with the aim of examining various interface operations. These included using the "Classify" button to activate the classification engine, generate *Pantun* predictions, and to generate recommended *Pantun* that related with the classified Malay *Pantun*, the "Clear" button to reset the text area form allowing user to re-enter new Malay *Pantun* input, the "Show Details" button to view the model confidence in classifying the *Pantun* in visualization pie chart, and the "Show Model Metrics" button to display overall model metrics including the wordcloud and bar chart visualization. The specific outcomes of the functionality testing are listed in the Table 4.3 as shown below.

Table 4.3 Functionality Test

Test Case	Features	Developer Date: 22/06/2025	Evaluator Date: 10/07/2025
1	Classify Button	/	/
2	Clear Button	/	/
3	*Pantun* Prediction	/	/
4	Show Details Button	/	/
5	Show Model Metrics Button	/	/
6	Related *Pantun* Recommendation	/	/

Every feature was rigorously tested by the system developer currently and external assessor from Universiti Teknologi MARA (UiTM) which are three different students for the functionality test. They tried all possible available function provided in the Malay *Pantun* theme classification prototype. This two-tiered testing procedure ensured evaluation objectivity and ensured that the functionality was reliable for a range of users. During testing, the "Classify" button consistently processed user inputs and produced classifications with the aid of the SVM. The output illustrated the expected *Pantun* theme with the aid of confidence score. One of the most impactful elements was the "Related *Pantun* Recommendation", which suggested other *Pantun* that had thematic structures in common with the classified entry. This feature significantly increases the platform's educational and cultural value by facilitating user exploration and providing more exposure to traditional Malay *Pantun*. The successful completion of all six test cases validated the interface's resilience and deployment readiness.

4.4 Discussion

The evaluation of the Malay *Pantun* theme classification system developed in this project, which uses a multiclass SVM algorithm with TF-IDF feature extraction, yielded an accuracy of 68.06%. Despite falling short of 80%, this result demonstrates a significant level of thematic classification capability for the six distinct *Pantun* theme used in this study. A number of comparisons with earlier related studies in literary classification and poetry genre detection were made.

The latest classification type of *Pantun* Using SVM found that the accuracy was 81.71% (Irmanda & Astriratma, 2020). Their dataset was simple, consisting of only 470 *Pantun* categorised into three main themes. This contributed to the model's higher discriminative accuracy because there were fewer boundaries between classifications. Another earlier relevant study, Poetry Classification Using Support Vector Machines, by Jamal et al. (2012), employed SVMs to categorise poetry themes from a corpus of 1,500 samples across ten categories. The SVMs had a comparatively poor accuracy of 58.44%. This is in line with the pattern of the present study, which indicates that increased theme diversity and dataset size may lead to classification issues.

In a different study, Pal & Patel (2020) classified Hindi poems using a range of machine learning techniques. The best accuracy, according to their findings, was achieved by Naïve Bayes (86%), SVM (83%), and Logistic Regression (84%). The linguistic structure of Hindi poetry or differences in text preprocessing techniques were the cause of this. This highlights a potential limitation of SVM in literary settings. Kaushal and Dutta (2023) conducted a more recent study on the classification of Punjabi poetry. Their analysis revealed that SVM outperformed other algorithms with an accuracy of 82.17%. However, it must be noted that the study focused on fewer categories than the current project and employed domain-specific feature engineering techniques, such as syllabic patterns and rhyme detection.

The accuracy of 68.06% for this project, in contrast, might seem low, but it need be considered in the context of its larger scope and the difficulties presented by the structure of the Malay *Pantun*. Malay *Pantun* frequently contains symbolism, metaphor, and indirect meaning, which makes automatic classification especially challenging, in contrast to datasets used in earlier studies. Complexity is increased by the inclusion of six unique and culturally rich *Pantun* themes, many of which share linguistic markers. Furthermore, the project relied on conventional TF-IDF representations, which might not

adequately capture the contextual richness of *Pantun* verses, in contrast to earlier research that frequently used smaller, more precisely defined datasets or improved preprocessing. Table 4.4 shows a summarized comparison of the finding from previous study.

Table 4.4 Discussion on Previous Research

No	Title	Year	Result	References
1	Klasifikasi Jenis *Pantun* dengan Metode Support Vector Machines	2020	81.71%	(Irmanda and Astriratma, 2020)
2	Poetry Classification Using Support Vector Machines	2012	58.44%	(Jamal et al., 2012)
3	Automatic Multiclass Document Classification of Hindi Poems using Machine Learning Techniques	2020	83%	(Pal & Patel, 2020)
4	Analysis of Performance Metrics for Classification of Punjabi Poetry using Machine Learning Techniques	2023	82.17%	(Kaushal & Dutta, 2023)

There are also experiment being conducted to compare the achieved result from self-developed SVM algorithm and also the optimized SVM and other algorithms such as the Naïve Bayes and K-Nearest Neighbour (KNN). The experiment were conducted using the RapidMiner tool which used the processed data. Among all of these algorithms, the SVM that is reside in the RapidMiner tool recorded the highest accuracy of 93.06% followed by Naïve Bayes which is 84.10%, then KNN which is 77.62% and finally the self-developed SVM which is 68.06% of accuracy. The result achieved from this experiment highlight the capability of optimized SVM produced highest accuracy which outperformed other algorithms. Table 4.5 shows the comparison of the Malay *Pantun* theme classification with other algorithms.

Table 4.5 Malay Pantun Theme Classification with Various Algorithm

No	Algorithm	Result
1	SVM(Rapid Miner)	93.06%
2	Naïve Bayes	84.10%
3	K-Nearest Neighbour	77.62%
4	Self-Developed SVM	68.06%

Other experiment were conducted to analyse the factor that affect the SVM's model performance. This experiment used the dataset of Malay *Pantun* that have three different themes which are "*Pantun Percintaan*", "*Pantun*

Kembara dan Perantauan", and "*Pantun Jenaka dan Permainan* ". It consists of 645 dataset which build from 215 *Pantun* from each themes. This experiment shows the highest accuracy is achieved by the SVM model with kernel RBF and data ratio 70:30 with accuracy of 75.26%. Overall, the model performance of Malay *Pantun* theme classifier has improved due to the bigger dataset size for each theme and the use of fewer themes, helped to reduce the thematic and linguistic complexity in traditional Malay *Pantun*. Table 4.6 shows the result of the experiment conducted with different dataset size and themes.

Table 4.6 Malay Pantun Theme Classification with Three Themes

Total Dataset	Data Splitting				Accuracy		
	Training		Testing		Kernel		
	Percentage	Number of Dataset	Percentage	Number of Dataset	Linear	RBF	Poly
645 *Pantun*	90%	648	10%	72	0.6769	0.7077	0.7385
	80%	576	20%	144	0.7054	0.7519	0.7519
	70%	504	30%	216	0.7216	0.7526	0.7371

Despite these challenges, the developed system demonstrates that SVM-based classification is possible for traditional Malay literature. The improved-on recall and per-class precision scores show that the system can consistently identify different themes. The primary focus of third objective which to evaluate the performance of the SVM algorithm for Malay Pantun theme classifier is achieved. By completing this objective, the project not only gain the evaluation of the SVM model's performance but also offers insightful information about its advantages and disadvantages.

4.5 Summary

The project has completed all three objectives that have been outlined in the first chapter. The identified problems and issues could be addressed and solve through the development of the Malay *Pantun* theme classifier prototype. Table 4.7 shows the project summary.

Table 4.7 Project Summary

BACKGROUND OF STUDY	PROBLEM STATEMENT	OBJECTIVE	SCOPE	LITERATURE REVIEW	METHODOLOGY	OUTCOME
POETRY CLASSIFICATION: Machine learning field, particularly NLP, enables automated poetry classification into its own distinguish themes. **MALAY PANTUN:** Cultural poetic form with unique structure but still have limited research and digital preservation efforts. (Jamal et al., 2012) **SVM ALGORITHM:** Design for text classification with high accuracy. It works by finding the optimal hyperplane which separates best different classes of data points and maximizing the margin between them(Irmanda & Astriratma 2020)	Improper labelled data reduce the effectiveness of classification to the most (Pal & Patel, 2020). Limited available dataset that comprise the Malay *Pantun* data into a single file(Ahmed et al., 2023) Lacks satisfactory study that are able to classify the Malay *Pantun* with high performance (Jamal et al., 2012).	To identify the implementation of SVM algorithm in text and poetry classification and its features.	TARGET USERS: Students and literary enthusiast DATASET: 720 Malay *Pantun* text of six different theme from Dewan Bahasa Pustaka(DBP) books called Kumpulan *Pantun* Melayu. ALGORITHM: Support Vector Machine(SVM) ATTRIBUTES: Four lines stanza of Malay *Pantun* and its corresponding labelled *Pantun* theme.	1.Implementation of SVM in various problem. 2.Implementation of SVM in poetry text classification.	Preliminary Analysis	A comprehensive understanding on the implementation of the SVM algorithm used in text classification related to poetry and its features were studied in seven related articles. A collection of 720 consolidated Malay *Pantun* dataset.
		To develop a prototype of Malay *Pantun* theme classifier using SVM.		1.Similar works related to poetry classification. 2.The implication of literature review.	Data Preprocessing	1.Processed Malay *Pantun* dataset. 2.Training and testing data.
					Design	1.System Architecture. 2.System Flowchart. 3.Prototype interface.
					Implementation	1.Malay *Pantun* theme classifier prototype. 2.Functional Malay *Pantun* theme classifier website.
		To evaluate the performance of the SVM algorithm for Malay *Pantun* theme classifier.		Types of evaluation metrics use to evaluate the model performance	Evaluation	Result of model performance using evaluation metrics: Accuracy: 0.6806 Precision: 0.6830 Recall : 0.6806 F1-Score : 0.6789

4.6 Conclusion

In conclusion, this chapter dive in a thorough examination of the results derived from the data collected using a Malay *Pantun* theme classification system. Two important elements of a thorough analysis that addressed the first project objective which are focus on to identify the implementation of the SVM algorithm in text and poetry classification and its features. It were conduct by a thorough review of relevant studies using Support Vector Machine (SVM) for literary classification tasks and a discussion of feature representation and dataset characteristics. This chapter also covered the results of the project's second objective, which was to develop a prototype of Malay *Pantun* theme classifier using SVM.

As we proceed to the third project objective which is to evaluate the performance of the SVM algorithm for Malay *Pantun* theme classifier. The results are examined with various experiment conduct to achieved the best performance. A confusion matrix was made first. The accuracy, precision, recall, and F1-score metrics were then used to analyse the model's performance. The overall model performance achieved was 68.06% of accuracy, with per-class scores showing strong performance for some themes, such as "*Pantun Budi*," and moderate results for others. The classification system operated reliably, and the 90-10 dataset split yielded the best outcome. By carefully examining the implementation of the SVM algorithm, developing a functional prototype for *Pantun* theme classification, and carefully assessing model performance, the project successfully achieved all of its objectives.

CHAPTER 5

CONCLUSION

This chapter provide a comprehensive review of the Malay Pantun Theme Classification project's development and results. The primary objective of the project was to develop a machine learning-based prototype that classifies Malay *Pantun* into thematic groups using the Support Vector Machine (SVM) algorithm. Every step of the development process, from data preprocessing and model training to evaluation, was guided by clearly defined objectives. The impact and execution of the project were successful overall. It has accomplished all of its objectives and set the stage for future research and advancements in the categorisation of literary and cultural works.

5.1 Summary of Project

To sum up, this project has successfully achieved its main objectives and significantly advanced the field of literary text classification, particularly in the area of Malay *Pantun* preservation and analysis using machine learning. Poetic and cultural texts can be processed and understood using structured, data-driven techniques, as demonstrated by the use of a compact and efficient SVM-based prototype. Throughout the development process, great attention was paid to the prototype's technical effectiveness and suitability for real-world educational or research applications. Methodical approaches were taken to each project goal, and the outcomes confirmed that using computational models in digital heritage projects is feasible. The following subsections summarise the progress made towards each goal.

5.1.1 Summary for Objective 1

The project's main objective is to identify the implementation of SVM algorithm in text and poetry classification and its features. This objective were accomplished through review of relevant literature and research that employs SVM for text classification tasks particularly those involving structured or poetic formats was carried out. The project examined the theoretical benefits of SVM, such as its ability to handle sparse and high-dimensional text features and its consistent performance in multi-class settings. These discoveries influenced the selection of the TF-IDF-based feature extraction method and influenced the development of the prototype. This study validated the SVM algorithm's suitability for language processing and provided a foundation for tailoring it to the particular characteristics of Malay *Pantun*. In doing so, the project contributes by contextualizing a well-established model for a traditional cultural dataset.

5.1.2 Summary for Objective 2

This project's second objective is to develop a prototype of Malay Pantun theme classifier using SVM. This was achieved by developing a fully functional prototype that processes *Pantun* text after completing several preprocessing steps, including text normalisation, tokenisation, stopword removal, stemming, and TF-IDF transformation. These features were then categorised into six preset *Pantun* themes using the SVM algorithm. The prototype's lightweight design ensures that it can be used with standard computer hardware, and its performance and usability were assessed. Its architecture promotes scalability, transparency, and future extensibility for teaching or research. Therefore, by transforming theoretical approaches into a practical tool that strengthen the bond between technology and literary tradition, the prototype offers a useful contribution.

5.1.3 Summary for Objective 3

The third objective of this project is to evaluate the performance of the SVM algorithm for Malay Pantun theme classifier. This was accomplished through a structured evaluation process that included model performance metrics such as accuracy, precision, recall, and F1-score, as well as a 90:10 training-testing split dataset. The results showed that the classifier performed consistently across all classes, indicating that the model generalises well, despite the linguistic ambiguity and thematic overlap that are commonly present in *Pantun*. The model performance evaluation also included a confusion matrix to visualise prediction patterns and errors and to offer opportunities for further analysis and improvement. This stage verified that the SVM classifier is suitable for poetic data and validated the preprocessing and feature extraction techniques used in the prototype. These findings contribute a meaningful performance baseline for future works on literary classification, particularly in Malay-language or culturally specific domains.

5.2 Project Contribution

This project contributes to computational linguistic and cultural heritage by focussing on the theme classification of traditional Malay *Pantun*. The first objective, which is to identify the implementation of SVM algorithm in text and poetry classification and its features, has been accomplished through a thorough study of SVM and its use in the classification of structured poetic texts. This study shows how well SVM performs in multi-class classification and how well it handles high-dimensional data. The findings address a justification for the result achieved in the literature artificial intelligence's under-representation in Malay literary fields, especially *Pantun*. This project demonstrates the algorithm's cultural relevance in addition to confirming its technical viability by applying it to a dataset that includes information that is culturally significant.

The development of a thorough and functional prototype supports the second objective, which is to develop a prototype of Malay Pantun theme classifier using SVM. The prototype successfully categorised *Pantun* into six thematic groups based on important linguistic features with the aid of a structured Natural Language Processing (NLP) pipeline. Students, educators, and cultural researchers will undoubtedly find this categorisation tool helpful as it simplifies the process of organising and assessing *Pantun*. By incorporating modern machine learning techniques into the study of Malay heritage, the prototype ensures that *Pantun* will remain searchable, accessible, and meaningful to younger generations.

The third objective, which was to evaluate the performance of the SVM algorithm for Malay Pantun theme classifier, was contribute to structured testing and performance evaluation using standard metrics. The prototype was evaluated using a training and testing split, and it consistently produced results in accuracy, precision, F1-score and recall. These results demonstrate that the classifier is capable of accurately interpreting and categorising *Pantun* across a range of themes. The evaluation also attests to the quality of the dataset, which was collected and arranged to make this work possible. The project provides a useful prototype, an organised dataset, and a reliable Malay *Pantun* theme classification model.

5.3 Project Limitation

Despite the fact that this project successfully produced a functional prototype for the classification of Malay *Pantun* themes using the SVM algorithm, several problems were encountered that affected the scope and findings of the study. The study's sole focus on classifying Malay *Pantun*, without any extension to other poetic forms like *Syair*, *Sajak*, or *Gurindam*, is a major flaw. The choice to focus solely on *Pantun* was made in order to ensure depth and specificity. Furthermore, the classification is limited to the thematic level alone, neglecting other poetical

components like structure, metaphor, or cultural symbolism that may also have an impact on interpretation.

Another additional limitation is the nature and organisation of the dataset that was utilised. The source of *Pantun* was published book from DBP which is "*Kumpulan Pantun Melayu*" (Bakar, 2018), which provided a well-chosen and structured collection for model training and development. Even though this ensures data consistency and quality, the model is not tested against more informal or varied *Pantun* sources, such as those written in regional dialects, mixed languages, or contemporary digital forms. The prototype therefore performs best with *Pantun* that is traditionally structured and may not perform as well with variations that are not. The model may not be as broadly applicable in more informal or modern literary contexts.

Additionally, the project's algorithmic scope was constrained. Only the SVM was implemented despite considering a variety of classification algorithms, primarily due to its proven effectiveness in text classification tasks and development time constraints. This limited investigation might have overlooked the relative performance of other modern algorithms, such as ensemble methods or deep learning models, which could result in higher accuracy or adaptability in future implementations. Furthermore, the system is designed to handle text-based inputs only and does not support other formats like speech, images, or handwritten. Finally, the prototype is designed to only accept well-structured, four-line *Pantun* that follows the standard format. This strict structural expectation makes the classifier unsuitable for handling non-standard *Pantun* forms. Within the given parameters, this ensures model accuracy and reliability while restricting the range of *Pantun* that can be processed.

Despite these drawbacks, the project provides a solid foundation for further study and advancement in the area of automated language literary

classification. This prototype also supports ongoing initiatives to conserve and promote Malay poetic heritage through computational methods. This research indirectly demonstrates the technical feasibility of the Malay *Pantun* theme classification using machine learning techniques specifically SVM as the main core. By offering this a structured approach this could be a meaningful contribution to the Malay literary field.

5.4 Project Recommendation

Subsequent research could look into extending the prototype by integrating machine learning algorithms other than SVM, such as random forests, decision trees, or even deep learning models like CNN and Transformers. Although the current prototype works well with a manually constructed SVM, looking into other models could increase classification accuracy and allow for better generalisation, particularly in cases where language or *Pantun* structure deviates from standard forms. Comparative analyses of different algorithms enable a deeper comprehension of the most effective techniques for literary classification in low-resource languages like Malay. The disadvantages of using individual classifiers may be reduced and robustness may be increased by using ensemble approaches. These advancements in *Pantun* theme classification systems increase future accuracy and adaptability.

Another area that requires attention is expanding the size of the dataset used for training and evaluation. Despite being well-organised and validated, the current dataset is limited to a single published source and strictly follows the conventional form of *Pantun*. Future projects should incorporate a wider range of sources, including digital platforms, oral traditions, regional dialects, and contemporary *Pantun* formats. Such inclusion led to more linguistic diversity and a more accurate portrayal of *Pantun*'s evolution in modern contexts. Furthermore, if the prototype could handle incomplete stanzas, mixed-language *Pantun*, or more creative poetic structures, the classifier would be more inclusive

and able to support broader cultural use cases. The quality and authenticity of enlarged datasets could be further improved by partnerships with regional cultural organisations or linguistic specialists.

Lastly, the prototype's delivery method and user interface can be further improved to increase accessibility and engagement. The classifier, currently in use as a prototype for academic testing, could be adapted to produce a mobile-based platform that would be more user-friendly for educators, cultural organisations, and students. Features like real-time *Pantun* submission, classification history, theme distribution visualisation, and integration with educational resources would make the system more practical in real-world situations. Additionally, implementing feedback mechanisms that let users confirm or modify theme predictions would encourage community participation in the preservation of Malay heritage while also improving model learning over time. The prototype would develop into a more important and dynamic digital heritage tool for upcoming enhancements.

5.5 Conclusion

In conclusion, this project has successfully achieved its primary objectives, which were to identify the implementation of SVM algorithm in text and poetry classification and its features, to develop a prototype of Malay Pantun theme classifier using SVM, and to evaluate the performance of the SVM algorithm for Malay Pantun theme classifier. The development process has highlighted the educational and cultural advantages of automating *Pantun* theme classification to demonstrate that SVM can be used for literary classification. With the creation of a working prototype and a validated dataset, the project has established a valuable foundation for further academic study and digital cultural preservation. These show the project’s visibility and accessibility of Malay *Pantun* in both scholarly and cultural contexts.

Despite the promising results of the prototype, several shortcomings have been identified including algorithmic diversity, dataset, and structural limitations. These challenges offer opportunities for further development, including exploring alternative algorithms, expanding the number of data sources, and enhancing the system's usability and adaptability. The recommendations are intended to act as a guide for further research into developing more inclusive and adaptable literary classification schemes. By addressing these issues, future work can build on the positive aspects of this project and overcome its present shortcomings. The project not only demonstrates the successful application of machine learning to traditional literature, but it also contributes to the preservation of the richness of Malay poetic heritage in a modern and accessible manner.

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